Platform Information Design and Competitive Price Targeting

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Abstract

We study platform information design, in which a platform provides information about individual consumers, enabling sellers to set targeted prices. Because targetable information partitions the market and might intensify seller competition, what information should be provided remains an empirical question. We collect novel micro data from Steam, the dominant PC video game platform, and use granular consumer purchase and usage data to estimate their heterogeneous demand for video games. We then develop an equilibrium targeted pricing model to simulate counterfactual outcomes under all possible information designs. We demonstrate that the profit-maximizing information design—which distinguishes heterogeneous consumer types while accounting for the design's equilibrium impact—yields substantially higher profits compared to demand-based heuristics that ignore the equilibrium impact.

Keywords: information design, competition, price targeting

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1 Introduction

A platform collects detailed data on individual consumer behavior across all its third-party sellers. These comprehensive data allow the platform to pin down each consumer's preference and price sensitivity, putting it at a uniquely powerful position: The platform can decide what information about each consumer to communicate to sellers and what to withhold from sellers, thereby influencing whether and how prices and promotions are targeted to individual or groups of consumers. In practice, a platform often designs of its infrastructure or rules to influence what information is used in targeting (Bhargava et al., 2020). For example, advertising platforms such as Google and Meta provide predefined interest labels about individual consumers, effectively dividing them into segments. Online retail platforms such as Amazon and eBay are also developing lists of consumer segments that sellers can target with price promotions or coupons.

Despite the increasing popularity of targeting, platforms do not have a clear understanding on how they should optimally design targetable segments. From Shopify's segmentation guidelines to Google's Topic API and Facebook's behavioral taxonomy, practitioners often segment consumers based on demographics and behavioral metrics, with the intent to group consumers with similar preferences into the same segment. Whereas these segment designs reflect the established academic insight that a *monopolist* can benefit from targeting consumer preferences (Rossi et al., 1996; Smith et al., 2023; Dubé and Misra, 2023), they rarely consider the implications on *competing* sellers' equilibrium strategies.

The equilibrium implications of price targeting are important because a segmentation scheme partitions consumers into submarkets—each with its own local market structure—and consequently affects competing sellers' prices and equilibrium outcomes. These equilibrium implications, however, are nuanced. Theoretical and limited empirical works have shown that the impact of competitive price targeting on profits and welfare are directionally ambiguous (Holmes, 1989; Corts, 1998; Besanko et al., 2003; Dubé et al., 2017). In addition, theoretical predictions depend on conditions that are difficult to verify empirically, and do not easily generalize to realistic platform markets. Moreover, beyond the binary case of whether or not firms can target, the extant literature is mostly silent about how targetable segments should be designed.

In this paper, we characterize the platform's design of targetable segments as an information design problem, in which the platform-designed segments reveal information about individual con-

sumer demand to sellers, who then use this information to set optimal targeted prices.[1] Using detailed micro data from Steam, the dominant PC video game platform, our counterfactual simulations demonstrate that incorporating equilibrium considerations is crucial for the optimal platform information design.

We assemble a novel dataset containing individual-level panel data on consumer purchase and usage behavior, and game characteristics derived from texts that describe the game's content and style. Using these data, we estimate a structural demand model that captures consumers' joint decisions of purchase and usage, measuring heterogeneous consumer preferences over multi-dimensional product characteristics. In particular, the model allows consumers' expected usage to affect their purchase decisions, thereby uncovering consumer preferences using data on which games they purchase and on how intensively they engage with these games. Our demand estimates identify eight latent consumer types, who differ in their overall engagement with gaming, tastes for specific game features, and sensitivities to prices. Importantly, assuming zero marginal costs, our estimates imply equilibrium uniform prices that closely match the observed uniform prices on Steam, lending credibility to these estimates as the basis of counterfactual equilibrium outcomes.

Next, we develop an equilibrium targeted pricing model to simulate alternative platform information design scenarios. We extend the standard uniform pricing equilibrium (Berry et al., 1995)—where sellers are informed about the consumer type distribution but have no information about individual consumer types—to a targeted pricing equilibrium where sellers also have some information that allow them to infer each consumer's type. When all sellers have the same information about each consumer, the equilibrium is one where the platform effectively groups consumers into segments, with sellers competing for each segment as a distinct submarket.[2] The model also generalizes to the case where sellers hold different information about each consumer and thereby compete in partially overlapping submarkets (see Appendix C).

Our results demonstrate that targeted pricing significantly improves total profits when implemented with the optimal information design—one that maximizes the total profits among the platform and all sellers. We arrive at this first conclusion by simulating all $4,140$ information designs, each representing a distinct information set. We find that targeting based on full information generates a 2.88% increase in total profits relative to uniform pricing, which is equivalent

[1] The term "information design" follows Bergemann and Morris (2019) in that the information designer (who is the platform) commits to the choice of the best information structure for a set of participants in the game (who are sellers that use this information for targeted pricing).

[2] In this case, one can interpret the platform's information design either as provision of information (which is our primary interpretation) or as a rule that restricts what information price targeting can be based on.

to approximately $124 million per year. However, targeting based on coarse consumer information generates drastically different market outcomes depending on how the information set is designed. For example, among all "Coarse 2" designs—allowing just two price segments—the platform can achieve profit increases over uniform pricing that range from 1.74% to a mere 0.05%, suggesting that optimal information design plays a critical role given the granularity of targetable information.

We proceed to demonstrate that equilibrium considerations are paramount in obtaining the optimal information design. We arrive at this second conclusion by simulating counterfactual designs of targetable segments based on the grouping of consumer types with similar price elasticity patterns, among other demand-based heuristics. We find that, holding fixed the level of granularity, these heuristics consistently lead to much less effective targeting than the optimal information design. For example, with two targetable segments, elasticity-based heuristics yield an approximately one percentage point lower profits compared to the optimal design, leaving $43 million in annual profits on the table. This profit gap persists across alternative heuristics and remains robust to different sources of consumer heterogeneity, demonstrating that demand-based heuristics, which represent similarity in consumers' price sensitivities or tastes, fail to approximate the platform's optimal design that also accounts for its impact on sellers' equilibrium strategies.

Beyond the maximization of total profits across the platform and all sellers, we also demonstrate that the optimal information design substantially improves consumer welfare, the platform's profit, and the balance of competition across large versus small sellers, suggesting that well-designed information structures can simultaneously benefit a broad set of market participants in the ecosystem.

Our findings offer important managerial implications for platforms that provide consumer information to sellers. The prior literature emphasizes how a single seller can use advanced models and rich datasets to uncover individual-level demand heterogeneity, enabling precise targeting of distinct consumer types. In contrast, we demonstrate that in a marketplace with competing sellers, the platform must also account for how its information design affects seller competition in equilibrium. This equilibrium consideration is especially crucial when the platform intends to provide only a limited number of consumer segments—a likely scenario given practical constraints such as implementation or menu costs (Sheshinski and Weiss, 1977), privacy concerns (Goldfarb and Tucker, 2019), and fairness considerations (Kahneman et al., 1986), all of which tend to favor the provision of few coarsened consumer segments. To this end, we develop an empirical framework that captures how the information available to competing sellers shapes equilibrium prices, profits, and consumer surplus. Our framework is generalizable across platforms with diverse consumer

compositions and objectives.

Related Literature. We extend the literature on competitive third-degree price discrimination by addressing the information design problem faced by online platforms. Prior theoretical work has shown that targeted pricing can either enhance or harm competing sellers' profits, depending on the structure of consumer preference heterogeneity (Thisse and Vives, 1988; Holmes, 1989; Shaffer and Zhang, 1995; Corts, 1998; Chen and Iyer, 2002; Armstrong, 2006). However, the direction of profit (and welfare) impacts relies on conditions that are not easy to verify empirically.[3] In addition, existing theoretical findings are typically derived under restrictive assumptions—most notably, markets with only two single-product sellers and limited forms of consumer heterogeneity—and do not easily generalize to modern marketplaces characterized by richer consumer heterogeneity and market structures. In the meantime, a small but growing empirical literature similarly focuses on settings with two competing firms and limited consumer heterogeneity (Besanko et al., 2003; Dubé et al., 2017). Our contribution is twofold. First, we extend the empirical literature to a more realistic platform setting involving numerous single- and multi-product sellers and multi-dimensional consumer heterogeneity. Second, we introduce a framework that enables simulations of counterfactual equilibrium outcomes beyond the boundary cases of full-information and no targeting to systematically evaluate the continuum of intermediate information designs—where platforms enable targeting based on partial information about consumers.[4]

Our paper also contributes to the growing literature on platform information design, particularly through the design of infrastructure and rules (Bhargava et al., 2020). Most existing studies on information design focus on providing information to consumers—about sellers or products—often through mechanisms such as reputation systems, warranties, or quality guarantees (Hui et al., 2021; Vatter, 2022; Hopenhayn and Saeedi, 2022; Hopenhayn et al., 2022; Hui et al., 2023). In contrast, we examine the platform's role in providing information to sellers, specifically shaping sellers' beliefs about individual consumers through the design of coarse segmentation. Our focus on competitive price targeting distinguishes this work from the literature on monetizing data through targeted advertising (Bergemann and Bonatti, 2024), and our conceptualization of information design differs from studies on the market for data transactions between sellers and intermediaries

[3]For example, Corts (1998)'s key condition is whether two competing firms' rank orders between a "strong" consumer type and a "weak" consumer type coincide, a condition defined on firms' best response functions between targeted-pricing and uniform-pricing equilibrium.

[4]While not being their main point, Dubé et al. (2017) also empirically examine how targeting based on behavioral types or geographic locations affects equilibrium outcomes, answering a question related to ours—which consumer information should sellers target.

(Pancras and Sudhir, 2007; Bergemann and Bonatti, 2015).

Broadly, our empirical study also presents new insights on the pricing of digital content, especially digital video games (Rao, 2015; Sunada, 2018; Brecko, 2023).

The rest of the paper is organized as follows. Section 2 conceptualizes the platform information design problem. Section 3 introduces the context and data. Section 4 sets up the demand and equilibrium pricing model. Section 5 presents estimation results. Section 6 discusses the value of information design through counterfactual simulations. Section 7 concludes the paper.

2 Conceptual model: Information and competitive price-targeting

Setup. We present a simple conceptual model to illustrate the importance of accounting for seller competition in a platform's information design. Assume that two single-product sellers, A and B, compete for each consumer, i. The consumer's utility for product $j \in \{A, B\}$ depends on her taste for the product, β_{ij}, a price coefficient normalized to -1, and a type-1 extreme value utility shock:

$$U_{ij} = \beta_{ij} - \text{Price}_{ij} + \epsilon_{ij}. \tag{1}$$

The consumer can also choose not to purchase, which yields $U_{i0} = \epsilon_{i0}$. Without loss of generality, we normalize both sellers' marginal costs to zero.

Consumer taste for each product, β_{ij}, is heterogeneous. Assume β_{ij} can take either a high value (H) or a low value (L), resulting in four types of consumers, labeled $1, 2, 3, 4$, who are characterized by preferences $(\beta_{iA}, \beta_{iB}) = (L, L)$, (L, H), (H, L), and (H, H), respectively. We set $H = 2$ and $L = 1$, and assume each type constitutes 25% of the market. We also assume that sellers know the value of H and L and the proportion of each consumer type in the market, but not individual consumer i's type. Therefore, without additional information, sellers set uniform equilibrium prices for all consumers based on the distribution of consumer types.

Information design. We assume the platform knows each *individual* consumer's type. Because the platform controls the information environment, it can decide what individual consumer information to communicate to sellers, thus allowing them to target groups of consumers using the information. Formally, we characterize a specific information design as a partition of the set of consumer types, $\{1, 2, 3, 4\}$, where each partition corresponds to a grouping of types into coarsened segments.

Given the platform's information design, sellers update their beliefs about each consumer's type based on the segment to which the consumer belongs, and set equilibrium prices accordingly. At one extreme, the platform can disclose full information about consumer types, designing an information set $\{\{1\},\{2\},\{3\},\{4\}\}$. In this case, sellers know and can set equilibrium targeted prices for each consumer type. At the other extreme, the platform can disclose no information about individual types, designing an information set $\{\{1,2,3,4\}\}$. In this case, sellers must rely solely on their prior beliefs about the consumer type distribution and set uniform equilibrium prices.

Alternatively, the platform can provide a coarsened information set which reveals certain aspect(s) of consumer heterogeneity and conceals other aspects. For example, the platform can provide information on consumers' taste for product B while withholding information about consumer taste for product A, designing the information set $\{\{1,3\},\{2,4\}\}$. In practice, this means grouping types (L,L) and (H,L) into coarse segment I, and types (L,H) and (H,H) into coarse segment II, and informing sellers of each segment's composition. Rational sellers will then infer that consumers in segment I are equally likely to be type 1 or type 3, with a similar inference for segment II. Sellers set equilibrium prices for each segment accordingly, following the first-order conditions detailed in Section 4.2.

Table 2.1: Seller A's Equilibrium Prices and Profits under Alternative Information Designs

		Uniform		Full-info		Optimal-Coarsened	
Type	Preference	Price	Profit	Price	Profit	Price	Profit
1	(L,L)	1.59	0.42	1.40	0.40	1.40	0.40
2	(L,H)	1.59	0.29	1.32	0.32	1.64	0.29
3	(H,L)	1.59	0.78	1.75	0.75	1.64	0.79
4	(H,H)	1.59	0.60	1.60	0.60	1.64	0.61
	Total($)		2.08		2.06		2.10
	Change		—		-1.0%		0.5%

Note: This table presents seller A's price and profit for each consumer type between the three information designs. By symmetry, seller B's price and profit are the mirror image (e.g., B's price for type 2 is \$1.75, the same as A's price for type 3). Across the information designs, "Uniform" refers to no targeting, "Full-info" refers to targeting each type, and "Coarsened" refers to when the platform provides the *optimal* coarsened information $\{\{1\},\{2,3,4\}\}$ that maximizes the sellers' and the platform's profits, allowing sellers to target two segments in total.

Equilibrium prices and profits. We solve for the equilibrium outcomes under all information designs in this conceptual example and report the resulting prices and profits for three information designs in Table 2.1. The first column presents the scenario in which the platform provides no

information about individual types (information set $\{\{1,2,3,4\}\}$), resulting in sellers setting uniform prices. The equilibrium price is \$1.59 for all consumers, which falls between the dollar-valued intercepts \$2 and \$1 for the two types. The second column presents the opposite scenario where the platform provides full information, $\{\{1\},\{2\},\{3\},\{4\}\}$. In this case, sellers set prices that target each consumer type. The way targeted prices differ from uniform prices reflects consumer heterogeneity. Specifically, both sellers lower the price to \$1.40 for type 1 consumers, who have low preferences for both products, and raise it to \$1.60 for type 4 consumers, who have high preferences for both products. In addition, seller A reduces the price for type 2 consumers, who prefer the competitor's product, and raises the price for type 3, who prefer the focal product. Seller B's prices are symmetric to A's.

One might expect targeting to increase profits. However, in this case, full-information targeted prices in fact reduces profits compared to uniform prices. As shown in Table 2.1, a decomposition of equilibrium profits by consumer type reveals an interesting pattern: with full information, seller A's profits from consumer types 1 and 3—those with lower preferences for seller B—decline by about 4%. The profit loss arises because seller B, who targets these consumer types with lower prices (\$1.40 for type 1 and \$1.32 for type 3), effectively lowers seller A's residual demand. By symmetry, the same happens to seller B for types 1 and 2. In other words, although each seller sets type-specific prices to maximize its own profit from each consumer type, the competitor's targeting can lower residual demand, intensify competition, and lead to lower total equilibrium profits. This finding under competing sellers sharply contrasts with the case with a monopoly seller, where price targeting never hurts profits (Smith et al., 2023; Dubé and Misra, 2023).

Profit-maximizing coarsened information design. We further examine the 13 intermediate cases where the platform limits the consumer information shared with sellers and thus allows only partial targeting. We refer to these cases as *coarsened* information designs and present their profit increase relative to the no-information scenario in Figure 2.1. The design that maximizes total seller profit is $\{\{1\},\{2,3,4\}\}$, which increases profits by 0.5% relative to the uniform equilibrium (1.5% relative to the full-information case). This optimal design separates consumers based on whether they exhibit a low probability of purchasing either product but pools all other consumers. As shown in Table 2.1, under full information, seller A significantly loses profit in its stronghold, type 3, when seller B can poach them with a lower price. By contrast, under optimal coarsened design, seller B sets a less aggressive price, catering to the coarsened segment $\{2,3,4\}$, leading

to higher residual demand and profits for the seller A. Along the same line, two other coarsened designs that pool consumers with distinct tastes also result in profit gains. Conversely, designs that separate these two consumer types into distinct segments significantly reduce sellers' profits, with the least profitable design reducing total profit by 1.5%.

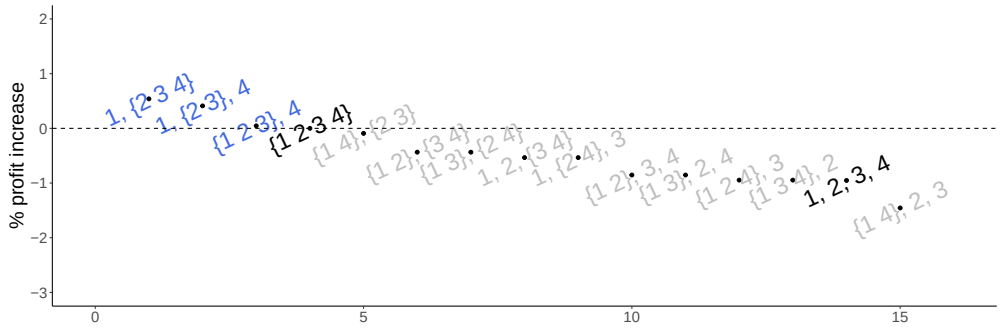


Figure 2.1: Information and total profits in equilibirum

Note: we present profit as the % increases relative to the uniform pricing equilibrium. For readability, we suppress the outer braces and braces for singleton sets. For example, we denote information set $\{\{1\},\{2,3,4\}\}$ by $1,\{2,3,4\}$.

Discussions. Our conceptual and structural model interprets the provision of segment labels as a source of information, which allows sellers—otherwise with no information about each consumer—to infer the consumer's type and set targeted prices. Alternatively, when the same information is imposed on all sellers, one can interpret the segment labels as a restriction on the aspects of consumer information that sellers are allowed to target. As a result, sellers set equilibrium prices for each segmented market, similar to zone pricing in physical retail (Adams and Williams, 2019). This alternative interpretation is isomorphic in our structural analysis and leads to the same findings.

Moreover, although we abstract away from the specifics of implementation, our proposed information design framework aligns well with current platform practices and can be supported by existing technologies. For instance, platforms like Google may assign consumer labels to indicate individual preferences, while platforms such as Amazon and eBay allow sellers to target pre-defined consumer groups through their seller interfaces (see Appendix Figure A.1). What may differ from existing practices is how to define targetable segments. Here, and in our empirical analysis, we demonstrate that optimal information design should account for its equilibrium impacts.

3 Context, Data, and Descriptive Statistics

3.1 Context: Steam

Steam, owned by Valve Software, is the world's largest digital marketplace for PC video games. It is estimated to hold 75% of the market share, offering over 30,000 games, software titles, and add-ons (Zuckerman, 2020). Consistent with this estimate, Google Trends data show that Steam consistently dominates consumer search interests, indicating limited competition from other platforms (see Appendix B).

Like other online marketplaces, Steam allows third-party sellers to list games and set their own prices. These sellers range from major publishers like Ubisoft and Sega to small indie studios with only a few titles. Consumers can access the platform for free but must purchase games to play them. Once purchased, a game is added to the user's library and remains accessible indefinitely. Steam processes all payment and charges a 30% commission fee from it. Since 2017, this commission has been Steam's primary revenue source, with third-party titles accounting for 98% of total transaction value and Valve's own games comprising just 2% during our sample period.

We highlight three industry details of Steam that help us study the information design question. First, because Steam processes all transactions and hosts all gaming sessions, it tracks the entire game purchases and usage history for each consumer. These micro data contain crucial information for us to estimate and understand consumer heterogeneity, a critical prerequisite for analyzing the profitability of price targeting.

Second, Steam does not allow targeted prices or targeted coupons during the sample period.[5] The absence of targeting allows us to observe the prices facing all consumers, thereby alleviating concerns about unobserved discounts that may arise due to targeting.

Third, several institutional features suggest that sellers primarily make pricing decisions to maximize game sales revenue. In particular, Steam does not sell advertising slots, so sellers do not make advertising decisions on Steam.[6] Additionally, although Steam recommends games based on individual consumers' interests, it has explicitly committed to not using prices in the recommen-

[5]This price uniformity is upheld with rare exceptions. Only one single anecdote exists where a large seller requested Steam to distribute targeted coupons to specific previous buyers, which is outside our sample period. Moreover, there is little anecdotal evidence of sellers circumventing the system by emailing consumers targeted coupons, primarily because Valve does not share consumers' email address with the publishers. See https://store.steampowered.com/privacy_agreement/ for more details.

[6]Steam's detailed policy regarding sellers' advertising and other marketing activities is publicly stated at `https://partner.steamgames.com/doc/marketing/tools`.

dation system.[7] This fact allows us to abstract away from modeling the effect of prices on Steam's recommendation and thus consumers' consideration sets. Moreover, upfront sales revenue remains the dominant source of income for sellers: only 3% of games offer in-game purchases, and in-game advertising is discouraged and has since been banned.[8] Further, because Steam distributes digital game copies via its own servers, sellers face negligible marginal costs (similar to other digital goods, Rao, 2015). Given these factors, it is plausible to assume that sellers set prices to maximize the revenue from game sales, thereby allowing for a tractable model to characterize sellers' equilibrium pricing behavior.

3.2 Data

Data collection. We collect consumer panel data directly from Steam's store website, community API, and storefront API. We start by randomly sampling one million existing Steam users in December 2016, with approximately 810,000 (81%) having accessible public accounts.[9] Subsequently, we narrow down the sample to 131,539 users who owned at least one game and logged in at least once in December 2016. For this sample, from February 5, 2017, to December 6, 2017, we track the games in each user's library and their cumulative playing time on a daily basis. We back out the purchase history by recording the day a user adds each game to her library, and we compute daily usage time at the minute level by taking the first difference on the cumulative time.[10] We retain a final sample of 34,548 active users who had positive spending and usage throughout the sample period and remained on Steam until at least November 2017.

Next, we collect daily price data for the 17,686 games available during the sample period. We observe both regular prices and the final prices post-discounts. Additionally, we collect the detailed characteristics of 4,581 popular games, such as developers, publishers, game descriptions, and user-generated tags, from a third-party data aggregator.[11]

[7] See https://steamcommunity.com/games/593110/announcements/detail/1612767708821405787 for details.

[8] On the current search page (accessed in October 2023), we count the number of games with user-generated tags "free to play" and "in-app purchases." We find that 6.1% of games (5,536/87,103) have the tag "free to play," and 2.7% (2,380/87,103) have the tag "in-app purchases." We verify that the tag "free to play" is accurate because the number of "free to play" games is similar to the number of games that currently have a price of zero. Furthermore, Steam formally banned in-game advertising in February 2025. See Steamworks Advertising Guidelines for more details.

[9] All Steam account profiles were public by default during that time. Users opting into the private mode or not registering on the Steam community would appear as missing or inaccessible accounts.

[10] A temporary data-collection glitch resulted in missing library-status records on 24 days spread across February, August, and September. We therefore exclude those days from the analyses. This omission has minimal impact on our structural estimation: it excludes 20% of usage days (58 of 302) and 2% of purchase weeks (1 of 43).

[11] The data from the third-party aggregator focuses on popular games on Steam as of 2020. Though the data only cover a subsample of games, it includes the most prominent games during our sample period, accounting for 99% of

Measuring product features by meta-tags. A crucial step in our analysis is to characterize product differentiation. Whereas each game possesses unique content and potentially appeals to a particular consumer type, it is computationally infeasible to uncover consumer tastes at the product level, especially with a large number of products.[12] To address this challenge, we characterize each game as a combination of key features that encapsulate its distinct content.

We derive these features from user-generated tags, which reflect each game's content in detail. For example, the popular first-person shooter (FPS) game, "Counter Strike: Global Offensive," has tags "FPS," "Shooter," "Multiplayer," "Competitive," "Team-based," "e-sports," "Tactical," and so on. These tags provide finer-grained information than genres and capture more nuanced differences across games that are relevant to users' purchase decisions.[13] However, given the 311 unique tags present across all games, estimating heterogeneous consumer preferences in such high dimensions is challenging. Therefore, we group tags with similar semantic meanings or describe similar contexts together to reflect the primary elements consumers seek in game content.

We use Global Vectors (GloVe) to capture the meaning of tags (Pennington et al., 2014). First, we obtain 200-dimensional word embeddings for each tag from a pre-trained GloVe model based on Twitter data.[14] Second, we cluster these word embeddings using a hierarchical clustering algorithm and manually correct misclassified tags. We then assign a "meta-tag" to each word cluster, summarizing the common feature described by the words. In total, we define nine meta-tags and present them in Appendix Table B.1. Each meta-tag is a binary variable indicating whether any original game tag belongs to the corresponding cluster. Note that these meta-tags are not mutually exclusive—a game can possess more than one meta-tag if it encompasses multiple features. On average, a game has 18 original tags and 2.6 meta-tags. In Section 4, we capture heterogeneous tastes for games using these defined meta-tags.

3.3 Descriptive statistics

We present descriptive statistics that motivate our subsequent analyses. First, we demonstrate that the market structure on Steam is complex, where a large number of sellers each offering one or

the usage time and 86% of the expenditure share. Games not covered are either too niche or have exited the market by 2020. For the game sample used in structural estimation, we collect the missing information for games that are not covered. We only have to fill in information for two out of the 171 focal games.

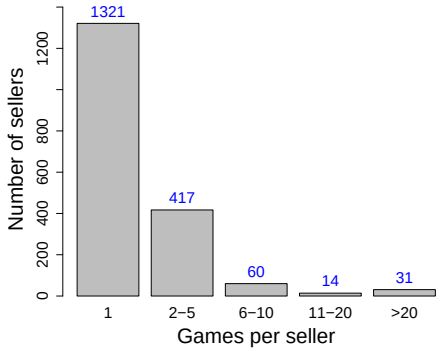
[12] With J products and R consumer types, one needs to estimate $J \times R$ parameters for product tastes, which grows quickly with J when R is high.

[13] See https://www.youtube.com/watch?v=Jsc7_XjFZ6U&t=319s and https://steamcommunity.com/discussions/forum/1/1480982971157417258/ for details.

[14] See *https://nlp.stanford.edu/projects/glove/* for details.

multiple differentiated products. Second, we document substantial consumer heterogeneity across multiple dimensions—such as purchase frequency, game-play intensity, and preferences for game features. Third, we examine current pricing patterns across games, highlighting the key source of identifying variation for demand.

(a) Number of games per seller (seller size)



(b) Revenue share by seller sizes

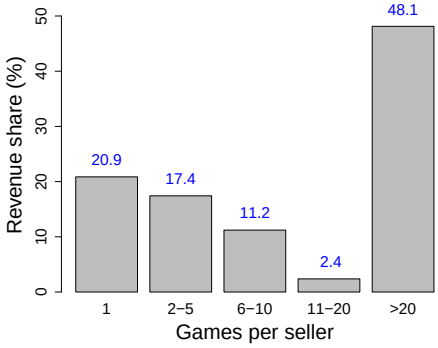


Figure 3.1: Market structure

Note: We use the 4,581 games for which we have detailed characteristics when computing the statistics. We group the 1,893 sellers in our data into five bins based on the number of games they own. Plot (a) describes how many sellers belong to each bin. Plot (b) illustrates the share of revenue generated by each bin. We use the gross revenue before the commission fee transfer in the calculation.

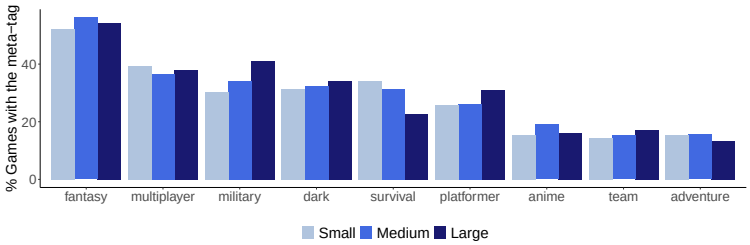
Market structure. We demonstrate that Steam's market structure differs significantly from canonical theory models of targeted pricing, which typically feature two firms that each offer a single product (Holmes, 1989; Corts, 1998). Figure 3.1 presents the distribution of seller sizes—in terms of their number of games—and revenue shares. We note that Steam is occupied by a few large, multi-product sellers and a long tail of small, single-product sellers. The 31 largest sellers, each with more than 20 games, collectively take 48% of the revenue. In contrast, 1,321 single-product sellers collectively contribute 21% of the revenue.

Moreover, because the large sellers often house multiple development teams that work concurrently on distinct projects, they not only offer a greater number of games but also span a broader range of product features.[15] As shown in Figure 3.2, most of the largest 31 sellers have some games that cover each of the product attributes, while smaller sellers are more dispersed in the game features developed. The presence of multi-product sellers and product differentiation within each seller brings forth a more complicated setting, one where existing theories do not apply. Thus,

[15] As an example, as of May 2021, Ubisoft employs over 20,000 individuals across more than 45 studios.

platform information design is an empirical question.[16]

(a) % of *games* with meta-tags: sellers of different size categories



(b) % of *sellers* with meta-tags: sellers of different size categories

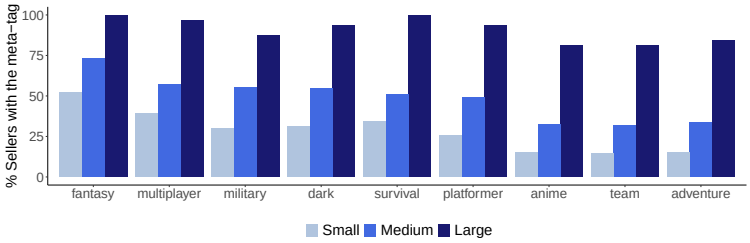


Figure 3.2: Larger sellers cover a wider set of features

Note: We categorize sellers with one game as "small," those with 2 to 20 games as "medium," and those with more than 20 games as "large." A seller has a certain meta-tag when at least one game of this seller has the meta-tag.

Consumer heterogeneity. We explore the behavior of active Steam users using the individual panel data and present the summary statistics in Table 3.1. Notably, consumers are highly engaged with Steam—over the ten-month sample period, a user logs into the platform for 44.8% days (about three times a week), spends $157, purchases 16 games, plays 17 games, and spends 1.5 hours playing games per day on average. The users are generally sensitive to price changes, with 41.1% purchases made at regular prices, 29.9% on discount, and 29.8% for free games on average. Users also display general preferences for games with certain meta-tags, such as multiplayer, military, and fantasy, over other meta-tags, such as team, adventure, or anime.

Moreover, the interquartile ranges in Table 3.1 reveal considerable heterogeneity among users in the three behavioral aspects—overall engagement, price sensitivity, and tastes in-game features. The overall engagement differs sharply across users. In the bottom quartile, consumers log in less than twice a week, spends $32 dollars, plays for 0.6 hour daily, purchases 4 games, and plays only

[16]Beyond the number of sellers and the presence of multi-product sellers, another departure from existing theory is the fact that the platform directly sells its own games while charging commission fees on other games. Nevertheless, as discussed in Section 3.1, platform-owned games take 2% of total sales revenue in the market, and its profits primarily come from third-party games.

Table 3.1: Heterogeneity across Active Steam Users

Statistic	Mean	St. Dev.	Pctl(25)	Median	Pctl(75)
% days logged in	44.8	23.5	25.8	43.0	62.7
spending ($)	157.1	257.1	32.0	80.0	182.4
N.games.purchased	15.5	25.7	4	9	18
N.games.played	16.5	16.6	6	12	21
daily hours	1.5	1.3	0.6	1.2	2.1
% regular price	41.1	27.5	21.1	37.1	55.6
% on discount	29.1	26.4	3.8	25.0	46.4
% for free	29.8	24.1	9.3	26.3	50.0
% multiplayer	55.1	26.7	37.0	54.5	72.7
% military	47.3	26.7	29.7	45.0	62.5
% survival	35.9	25.5	18.9	33.3	50.0
% team	29.0	25.3	11.4	25.0	40.0
% fantasy	47.7	26.1	33.3	50.0	64.9
% platformer	34.1	24.5	18.2	33.3	50.0
% dark	39.3	24.7	23.8	39.3	51.9
% adventure	12.7	16.3	0.0	9.1	20.0
% anime	10.3	14.0	0.0	6.2	16.2

Note: the table summarizes the distribution of three sets of behavioral statistics across the 34,548 active Steam users. The first set describes how engaged a user is, which includes the fraction of log-in days (when a user either purchases or plays something), the dollar amount spent, the number of games purchased and played, and the average playing hours per day. The second set presents the fraction of game purchases at regular prices, discounts, and when games are free. The third set shows the fraction of game purchases with certain meta tags.

6 games during the sample period. In contrast, users in the top quartile resemble heavy users, as they log in more frequently, spend more money and time, and purchase and play a larger set of games. In addition, price sensitivities and horizontal preferences across game features are also heterogeneous, as evidenced by consumers' purchases on games on discount and games with various meta-tags. For example, in the bottom quartile, only 21% of purchases are under the regular price; at the top quartile, this ratio rises to 56%, suggesting that some consumers are less price-sensitive than others. All of the systematic consumer behavioral differences signify substantial consumer heterogeneity, emphasizing the potential scope for price targeting.

Current pricing. Our primary source of price variation comes from the temporal price changes within each game. On Steam, sellers adjust prices by altering regular prices or setting temporary discounts.[17] For each game, we analyze how regular prices and final prices (i.e., effective prices after discounts) change over time and present the pricing patterns across games in Table 3.2.

We highlight two noteworthy observations. First, regular prices for most games rarely change. This observation differs from the console video game market, where regular prices tend to decline as

[17]See `https://partner.steamgames.com/doc/marketing/discounts` for Steam's rules on discounts' frequency, duration, and depth.

Table 3.2: Pricing pattern across games

Statistic	N	Mean	Min	Pctl(25)	Median	Pctl(75)	Max
average regular price ($)	17,686	8.7	0.0	3.0	6.1	10.0	100.0
SD(regular price) ($)	17,686	0.3	0.0	0.0	0.0	0.0	103.8
SD(final price)	17,686	1.4	0.0	0.2	0.9	2.0	46.3
% days on discount	17,686	11.0	0.0	4.3	9.2	15.5	100.0
average $ off on discount	14,005	4.9	0.1	1.6	3.6	6.6	78.9
average % off on discount	14,005	50.3	4.2	33.3	50.1	70.1	100.0
% during sales event days	14,005	32.3	0.0	0.0	31.7	50.0	100.0

Note: For each game, we calculate four statistics to describe its cross-time price variation: standard deviation of the regular price, standard deviation of the final price, number of days on discount, and the average percentage off when on discount. We present the distribution of these four statistics across games in this table.

used copies flood the market (Ishihara and Ching, 2019) or where sellers engage in price skimming (Nair, 2007). Second, final prices fluctuate more, primarily driven by occasional steep discounts. During our sample period, on average, a game is on discount 11% of the time, with an average discount depth of 50% off the regular price. About 32% of the discounts coincide with Steam's platform-wide sales event, and the rest of the discounts occur outside the sales event window throughout the year. Appendix Figure B.5 presents the pricing pattern of a game as a visual representation.

We leverage sharp price discounts to identify price sensitivities.[18] By combining the price panel with the consumer panel of purchases, we identify consumers' heterogeneous price sensitivities from how individual demand co-move with prices within short time windows. We present further details on pricing patterns and discussions about the identification in Appendix B.4.

4 Structural Model

4.1 Demand: consumer purchase and usage decisions

Overview. We model consumers' joint purchase and usage behavior using a two-stage framework. Similar to Crawford and Yurukoglu (2012), our model links purchase decisions to expected usage by assuming consumers are willing to pay more for games that they anticipate using more frequently.

[18] We abstract away from price variations across geographic regions (e.g., U.S. or Europe), as we solely observe price series in U.S. stores. Nevertheless, we collect cross-region prices for all games included for structural estimation. The timing and depth of discounts are highly aligned across geographic regions, with a median cross-region weekly price correlation of 0.90, alleviating concern about the impact of cross-region price differences on our estimates. We also abstract away from bundling, as it requires modeling consumer purchases over a bundle of products and increases the computational burden. An examination of available bundles during our estimation period revealed that this practice was not prevalent among games in our estimation sample.

In the data, purchases are less frequent than usage due to the durable nature of video games. As such, jointly modeling the two decisions enables us to leverage both decisions to capture consumer heterogeneity.

We illustrate the timing of a consumer's two-stage decision in Figure 4.1. The consumer starts a *week* with a set of games she already owns, which we refer to as her "library." She faces two purchase occasions, each of which allows her to purchase a game from the set of available (but not yet purchased) games and expand her library. Subsequently, on each *day* of the week, the consumer faces two usage occasions, each of which she can decide which game in her library to play and for how long. Figure 4.1 shows an example: with Games 1 and 2 in her library at the beginning of the week, the consumer considers buying Games 3, 4, both, or neither. If she purchases Game 3, she will decide how to allocate time among Games 1, 2, and 3 or other activities every day of the week.

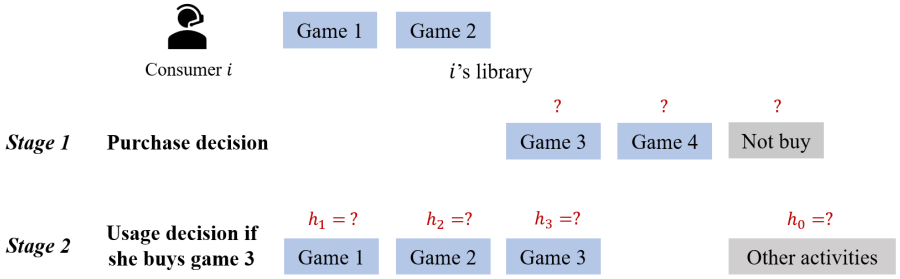


Figure 4.1: An example of consumer's purchase and usage decisions

Note: This figure describes the decision process of consumer i by an example. In the purchase stage, i chooses among games she does not own, or she does not buy anything. If i purchases a game, such as game 3, she will add it to her library. In the usage stage, she allocates time across games in her library and other activities.

Usage stage. We now characterize consumer decisions, starting with usage and then moving backwards to purchase. Let i be a consumer, j be a game, t be a usage occasion (there are two independent usage occasions per day). Let L_{it} be the consumer's library on usage occasion t, *including* the game she just purchased.

For usage occasion t, the consumer allocates her time across video games and other activities, maximizing her direct utility from leisure activities,

$$\begin{aligned} &\max_{\boldsymbol{h}_{it}} \Big\{ \sum_{j \in L_{it}} U_{ijt}(h_{ijt}) + c_t h_{i0t} \Big\}, \\ &\quad s.t. \sum_{j \in L_{it}} h_{ijt} + h_{i0t} \leq T^{max},\ h_{ijt} \geq 0, \text{ and } h_{i0t} > 0. \end{aligned} \tag{2}$$

In the above, U_{ijt} is the utility to play game j for h_{ijt} hours, a function of the consumer's persistent preferences and utility shocks. $c_t h_{i0t}$ is the consumer's utility from spending time h_{i0t} on non-gaming activities, from which she derives utility of c_t per hour. The time allocation problem has three constraints: a time-budget constraint, a non-negativity constraint, and a positive outside good constraint (one cannot spend all day gaming). The first two constraints are naturally satisfied given any time allocations. The third constraint holds for more than 99% of all consumer-day level observations.

Given these constraints, the usage decision in occasion t can be decomposed into two sequential stages: first, the consumer chooses which game to play; second, she decides how much time to allocate to the chosen game. We parameterize the utility function based on the multiple discrete-continuous (MDC) utility specifications in Bhat (2018).

$$U_{ijt}(h_{ijt}) = \gamma \cdot (\Psi_{ijt} + \mu_{ijt}) \cdot \log\left(\frac{h_{ijt}}{\gamma} + 1\right) + \omega_{ijt}, \tag{3}$$

where Ψ_{ijt} shifts the marginal utility of playing game j, and γ captures how quickly consumers are satiated from playing a game. Two separate utility shocks affect consumers at different points during the decision process: ω_{ijt} realizes before the consumer picks a game, capturing any factors influencing which game she chooses, and follows the type-1 extreme value distribution; μ_{ijt} realizes after the consumer starts playing the game, representing in-game stimuli such as game achievements, and follows a standard normal distribution.

A consumer's optimization problem goes as follows. First, the consumer chooses whether to play a game and, if so, which game to play, solving

$$\max_{j \in L_{it} \cup \{0\}} \left\{ \mathbb{E}_{\mu_{ijt}} \left[\gamma \cdot (\Psi_{ijt} + \mu_{ijt}) \cdot \log\left(\frac{h^*_{ijt}}{\gamma} + 1\right) + c_t h^*_{i0t} \right] + \omega_{ijt} \right\}, \tag{4}$$

subject to the budget and non-negativity constraints. The consumer expects herself to choose the optimal time allocation, h^*, between j and the outside activity, given any μ values during the session. Therefore, she makes the decision by taking an expectation over the distribution of μ_{ijt}.

Next, given the choice of j and the realized μ_{ijt}, the consumer decides how long to play game j. Substituting in the time-budget constraint,[19] the consumer maximizes the indirect utility function

$$\max_{h_{ijt}} \left\{ \gamma \cdot (\Psi_{ijt} + \mu_{ijt}) \cdot \log\left(\frac{h_{ijt}}{\gamma} + 1\right) - c_t h_{ijt} \right\}, \; s.t.\, h_{ijt} \geq 0 \,\text{and}\, h_{i0t} > 0. \tag{5}$$

[19]The constant term T^{max} drops out after the substitution.

The problem has a closed-form solution, $h^*_{ijt} = \max\{\gamma \cdot (\frac{\Psi_{ijt}+\mu_{ijt}}{c_t} - 1), 0\}$.

Purchase stage. Consumer i has two purchase occasions at the beginning of each week, and we denote a week–purchase-occasion combination t_w. We assume the consumer explicitly accounts for the changes in game library, and the associated usage decisions and payoffs, in the week following a purchase decision. That is, whether to purchase a game depends on how the expected maximum future payoff in the following week changes with the addition of the game. However, to maintain model tractability, we do not explicitly characterize the consumer's expectations of one purchase's influence on the other purchase occasions or subsequent usages beyond the week. Instead, we capture these values (and other potential drivers of purchase beyond the week's usage) with additional terms in the utility function. Formally, the consumer chooses among the outside goods and the available games she has not owned ($j \in \{0\} \cup J_{t_w} \backslash L_{it_w-1}$), solving

$$\max_{\{j \in \{0\} \cup J_{t_w} \backslash L_{it_w-1}\}} \left\{ \underbrace{\mathbb{E}\left[V^{usage}_{it_w}(L_{it_w-1} \cup \{j\})\right]}_{\text{expected usage value}} + \underbrace{V^{purch}_{ijt_w}}_{\text{purchase value}} - \underbrace{\alpha_{ij}\text{Price}_{jt_w}}_{\text{disutil. from expd.}} + \underbrace{\epsilon_{ijt_w}}_{\text{util. shock}} \right\}. \quad (6)$$

In the above, $V^{usage}_{it_w}$ denotes the maximum expected utility the consumer derives from playing games in the following week, given her library $L_{it_w-1} \cup j$, where the expectation is taken over the distribution of ω. $V^{purch}_{ijt_w}$ captures utility directly derived from purchasing game j, beyond the utility from playing the game. $\alpha_{ij}\text{Price}_{jt_w}$ captures the disutility from paying for the purchase, and ϵ_{ijt_w} follows Type I extreme value distribution and captures idiosyncratic factors influencing game purchase decisions.

Parameterization. In Table 4.1, we present how we parameterize consumers' usage utility Ψ_{ijt}, opportunity cost of time c_t, purchase utility $V^{purch}_{ijt_w}$, and price coefficients α_{ij}.

Table 4.1: Component specifications

usage	Ψ_{ijt}	$= \eta_j + \boldsymbol{X_j \lambda_{r(i)}}$
	c_t	$= \exp(c_{wkd} \cdot \text{weekend}_t + c_{smr} \cdot \text{summer}_t + c_{hld} \cdot \text{holiday}_t)$
purchase	$V^{purch}_{ijt_w}$	$= \delta_j + \boldsymbol{X_j \beta_{r(i)}} + \xi_{smr} \cdot \text{before sales event}_{t_w} + \xi_{hld} \cdot \text{sales event}_{t_w}$
	α_{ij}	$= \boldsymbol{\alpha_{r(i)} X'_j}$

Note: $r(i)$ indicates the latent type of consumer i. X_j is the game features (i.e., meta-tags), which we construct as described in Section 3. c_t is normalized to be 1 for weekdays that are neither during summer times nor holidays.

We include game fixed effects to account for differences in game popularity driven by unique

content and production quality. To allow for the possibility that consumers perceive the same game differently at the purchase and usage stages, we introduce two separate sets of fixed effects: δ_j in the purchase utility function $V_{ijt_w}^{purch}$, and η_j in the usage utility function Ψ_{ijt}. We also incorporate time effects to control for common temporal factors affecting consumer behavior: we allow c_t to vary across weekends, summer, and holidays to capture variations in the opportunity cost of playing; we let $V_{ijt_w}^{purch}$ differ during sales events and the two weeks leading up to them, capturing changes in consumers' propensity to purchase during and before the promotional periods.

We then use a latent-class model to capture unobserved heterogeneity across consumers, allowing arbitrary correlation among the heterogeneous parameters (Kamakura and Russell, 1989). Specifically, we denote $r = 1, 2, ..., R$ as discrete consumer types in the market and and ψ_r as the population share of type r. We allow for persistent heterogeneity across consumers in their overall engagement with gaming, preferences for game features, and price sensitivities. In Table 4.1, $\boldsymbol{X_j}$ includes a common intercept and the set of meta-tags representing game features, as constructed in Section 3. The parameter vectors $\boldsymbol{\lambda_{r(i)}}$ and $\boldsymbol{\beta_{r(i)}}$ capture the preferences of consumer type $r(i)$ for gaming in general and for specific game features in the usage and purchase stages, respectively. We also allow price sensitivities, $\boldsymbol{\alpha_{r(i)}}$, to vary by consumer type and interact with game features.[20]

Identification. Let $\boldsymbol{\Theta}_1 = \{\boldsymbol{\eta}, \gamma, c_{wkd}, c_{smr}, c_{hld}, \boldsymbol{\delta}, \xi_{smr}, \xi_{hld}, \sigma\}$ denote the set of homogeneous parameters, and let $\boldsymbol{\Theta}_2 = \{\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\lambda}, \boldsymbol{\psi}\}$ denote the set of heterogeneous parameters. We now provide an informal discussion on identification.

We identify the usage-stage parameters $\{\boldsymbol{\eta}, \boldsymbol{\lambda}, \gamma, c_{wkd}, c_{smr}, c_{hld}\}$ from consumers' choices of what to play and how long to play. Specifically, γ is identified by the typical duration of game-play across all titles, while η_j and $\boldsymbol{\lambda}_r$ are identified through variation in the frequency and duration of game-play across games with distinct features. In contrast, we identify purchase-stage parameters δ_j and $\boldsymbol{\beta_r}$'s from the deviations between consumers' purchase decisions and their expected usage values. For example, a consumer who frequently purchases games she rarely plays likely places high value on certain game features at the time of purchase. Price sensitivities are identified by consumers' responses to sharp discounts, which are the predominant source of price variation in the data. Temporal variation in consumer behavior helps identify the time shifters—weekend, summer,

[20] Allowing the price sensitivities to vary by game features fits the data considerably better. A plausible explanation is that, due to the recommendation system, consumers may be more frequently exposed to certain games than others. Games not surfaced to a consumer may not enter her consideration set, thereby reducing her responsiveness to discounts on those games. Because our primary focus is not on why consumers react to certain price discounts more than others, and because prices do not enter the recommendation system (Section 3.1), we take the different price responsiveness as given and capture it in a descriptive way.

and holiday effects for usage, and sales event effects for purchases.

Although we cannot identify the exact number of latent types in the market, we gradually increase the value of R and compare the corresponding model fit. We set $R = 8$ because it provides a good fit to the observed behavior while keeping the computational burden manageable, as later discussed in Section 5. Given R, we identify latent consumer types leveraging the long panel structure of our micro data, in which we observe, per consumer, 43 weeks of purchases and 244 days of usage, and up to two purchase or usage occasions per week or day.

Estimation. We estimate the demand parameters using maximum likelihood estimation. The individual likelihood function corresponds to the probability of observing the sequences of purchase and usage decisions, as shown in equation (7). From researchers' perspective, conditional on $\boldsymbol{\Theta_1}$, the randomness in consumer choices arises from two sources: the unobserved consumer type $\boldsymbol{\theta_{r(i)}} = \{\boldsymbol{\beta}_r, \boldsymbol{\alpha}_r, \boldsymbol{\lambda}_r\}$, and the unobserved demand shocks $\{\boldsymbol{\epsilon}_{t_w}, \boldsymbol{\omega_t}, \boldsymbol{\mu_t}\}$. Therefore, we first derive the probability conditional on $\boldsymbol{\Theta}_r = \{\boldsymbol{\Theta_1} \cup \boldsymbol{\theta}_r(i)\}$, and then integrate over the distribution of types. Because of the independence between $\boldsymbol{\epsilon}_{t_w}$ and $\{\boldsymbol{\omega_t}, \boldsymbol{\mu_t}\}$, we can separately compute the likelihood components for the sequences of purchase and usage decisions. The joint log likelihood, integrated across consumer types, is:

$$LL = \sum_i \log(L_i) = \sum_i \log \left(\sum_r \underbrace{\psi_r}_{\text{type share}} \cdot \underbrace{\prod_{t_w} \Pr(\text{purch}_{it_w}|\boldsymbol{\Theta}_r)}_{\text{prob. purchase sequences}} \cdot \underbrace{\prod_t \Pr(\text{usage}_{it}|\boldsymbol{\Theta}_r)}_{\text{prob. usage sequences}} \right). \tag{7}$$

In the above, type-specific likelihoods for purchase and usage at a given choice occasion are:

$$\begin{aligned} \Pr(\text{purch}_{it_w}|i, \boldsymbol{\Theta}_r) &= \frac{\exp(\bar{V}_{ij,r})}{\sum_{k \in J} \exp(\bar{V}_{ik,r})}, \\ \text{where } \bar{V}_{ij,r} &= \mathbb{E}[V^{usage}_{it_w}(L_{it_w-1} \cup j|\boldsymbol{\Theta}_r)] + V^{purch}_{ijt_w} - \alpha_{rj}\text{Price}_{jt_w} \end{aligned} \tag{8}$$

and

$$\Pr(\text{usage}_{it}|\boldsymbol{\Theta}_r) = \begin{cases} \Pr(0|i, \boldsymbol{\Theta}_r)^{\text{start}} + \sum_{j \in L_{it}} \Pr(j|i, \boldsymbol{\Theta}_r)^{\text{start}} \cdot \Pr(h_{ijt} = 0|i, \boldsymbol{\Theta}_r), & \text{if no positive hours,} \\ \Pr(j|i, \boldsymbol{\Theta}_r)^{\text{start}} \cdot f(h_{ijt}|i, \boldsymbol{\Theta}_r), & \text{if } h_{ijt} > 0, \end{cases} \tag{9}$$

For the usage stage, we separate the probability calculation based on whether usage is zero or positive. Specifically, we account for two sources of zero usage: (i) the consumer chooses not to start any game, or (ii) the consumer starts a game but allocates zero hours to play after observing the in-game shock μ. Given the optimal hours decision, $h^* = \max\{(\frac{\Psi+\mu}{c} - 1) \cdot \gamma, 0\})$, we map observed hour choices back to the distributions of $\{\boldsymbol{\omega}, \boldsymbol{\mu}\}$ and derive the probability of zero usage in equation (11), along with the conditional density for positive usage, $f(h_{ijt} \mid i, \boldsymbol{\Theta}_r)$, in equation (10).

$$f(h_{ijt}|i, \boldsymbol{\Theta}_r) = f\big(\mu_{ijt}(h_{ijt})\big) \cdot \frac{\partial \mu_{ijt}(h_{ijt})}{\partial h_{ijt}} = \phi\big((\frac{h_{ijt}}{\gamma} + 1) \cdot c_t - \Psi_{ijt}\big) \cdot \frac{c_t}{\gamma}, \tag{10}$$

$$\Pr(h_{ijt} = 0|i, \boldsymbol{\Theta}_r) = \Phi\big(c_t - \Psi_{ijt}\big) \tag{11}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ indicate the probability density function and the cumulative distribution function of the standard normal distribution. As for the decision to start a game, we map the observed choices back to the probability space of $\boldsymbol{\omega}$, which gives, for $j \in L_{it}$:

$$\Pr\left(j|i, \boldsymbol{\Theta}_r\right)^{\text{start}} = \frac{\exp\left(\mathbb{E}_\mu\left[U_{ijt}(h^*_{ijt})|j, i\right]\right)}{1 + \sum_{j' \in L_{it}} \exp\left(\mathbb{E}_\mu\left[U_{ijt}(h^*_{ijt})|j', i\right]\right)}, \tag{12}$$

and $\Pr\left(0|i, \boldsymbol{\Theta}_r\right)^{\text{start}} = 1 - \sum_{j \in L_{it}} \Pr\left(j|i, \boldsymbol{\Theta}_r\right)$. One can plug (10), (11) and (12) back into (9) to get $\Pr(\text{usage}_{it}|\boldsymbol{\Theta}_r)$. We use numerical integration to calculate the expectation over μ.

Due to computational constraints, we cannot estimate the model on our full dataset. Instead, we construct a subsample that focuses on popular games during the sample period. Specifically, we select 171 focal games that constitute a substantial share of revenue or usage time and randomly sample 6,314 active consumers. The game sample captures the key fraction of the market, with 9 games from Valve, 72 games from 16 large third-party publishers, and 90 games from 88 small third-party studios. We verify that our feature construction (Section 3) captures the diversified content of the focal games: on average, each title carries 25 original tags and 4 meta-tags, and across the 171 games, there are 97 unique meta-tag combinations. See Appendix D.1 for details on the sampling process.

We aggregate all other games into a "fringe game." We abstract away from the game features for the fringe games, normalize their prices to be 0, and introduce (i) a fringe-game fixed effect and (ii) random coefficients on the fringe game at both the purchase and usage stages. These

terms allow us to capture heterogeneity in consumers' tendencies to buy or play fringe games while preserving model traceability.

Discussions. We now discuss the simplifying assumptions we have imposed to maintain model tractability. First, we assume the consumer has two purchase occasions per week and two usage occasions per day. The assumption is consistent with the vast majority of consumers' choices in the structural sample: 97% of the consumer-week observations have two or fewer game purchases (95% conditional on at least one purchase), and 97% of the consumer-day observations involve playing two or fewer games (92% conditional on positive play time).[21]

Second, we assume each purchase decision explicitly accounts for its impact on usage only in the following week. This assumption rules out dynamic purchase decisions, such as stockpiling, and thus maintains a tractable, static model. This assumption is plausible given the short gap between a consumer's purchase time and her initial playing session: of all observations of consumer purchase and initial game-play, only 11% start more than seven days after purchase. If stockpiling were prevalent, one would expect a larger share of purchases to exhibit longer delays before initial usage. In addition, stockpiling would imply intertemporal substitution around major sales events, as consumers might delay purchases in anticipation of substantial discounts. However, as shown in Appendix B, estimated price sensitivities remain similar regardless of whether we control for the timing of platform-wide sales events. Moreover, post-estimation analysis reveals no evidence of a dip in the purchase shifter prior to sales events.

Third, we assume consumers' initial game libraries as exogenously given. One may be concerned that game libraries are selected based on consumers' unobserved heterogeneity. However, explicitly modeling these initial libraries is computationally infeasible.[22] Nevertheless, we show that most consumers own a small fraction of the focal set of games, so the initial libraries do not vary significantly across consumers to cause a meaningful selection bias. On average, consumers own 13 of the 171 focal games (median = 10; IQR = 5–18) before the sample starts. In Appendix C.1,

[21] We do impose that the two purchase occasions are independent, implicitly assuming that when making the first purchase in a week, the consumer does not account for the possibility that she may buy another game in the same week. This simplifying assumption avoids modeling dynamic purchase decisions and makes the model tractable. We note that only in 3% of the weekly observations does the consumer purchase two games, suggesting that this assumption is not crucial. Moreover, these stats hold across the full sample of active Steam users and titles, confirming that the independence assumption is consistent with typical behavior on Steam.

[22] It is computationally infeasible for two reasons. First, the set of games is large, leading to a large number of possible libraries (as the combination of owned games) and even larger number of possible paths to reach a given library. Second, because libraries evolve non-stationarily, standard approaches for imposing a stationary initial-condition distribution (e.g., Simonov et al. 2020) do not apply.

we conduct Monte Carlo simulations and show that demand parameters can be recovered when a similar fraction of games are purchased before the estimation sample begins. As such, we believe that assuming exogenous libraries does not lead to a significant bias in our demand estimates.

Fourth, we assume the consumer makes informed purchase decisions, abstracting away from uncertainty about game quality or match values. We consider this assumption plausible, due to the abundant information available at the time of purchase, including extensive user reviews on Steam, gameplay videos on YouTube (Li et al., 2025), and live streams from game influencers (Huang and Morozov, 2025). Furthermore, despite Steam's generous return policy, only 3.8% of the initial purchases are returned in our sample. The lack of return suggests that consumers have limited uncertainty about game quality at the time of purchase, and thus have no regret after they begin playing the game.[23] Nevertheless, we do allow for discrepancies between purchase- and usage-stage payoffs, and as we discuss later, these additional terms could capture some extent of information differences. We also argue that consumers are unlikely to learn about their own match value to game features over time, because consumers in our sample are experienced gamers. In our sample, the average consumer has been a player on Steam for 4.3 years (median 3.6 years) prior to the sample period, and therefore likely know their own preferences over different kinds of games. Consistent with this conjecture, the distributions of the features among consumers' first five and last five purchases are remarkably similar (see Appendix Figure B.3).

Finally, a variety of other factors may also influence consumer demand, including utility directly from ownership, advertising outside of Steam, personalized recommendations from Steam, learning of game content post-purchase, and future usage utility beyond the one-week horizon. Abaluck et al. (2022) argue that if the goal is to characterize equilibrium outcomes under counterfactual scenarios when demand is held fixed, one can model demand flexibly rather than separating preferences from alternative demand-driving mechanisms. In our case, we lack data to directly measure these channels. So, instead of modeling these channels explicitly, we capture them by parameters δ_j, $\boldsymbol{\lambda_r}$, and $\boldsymbol{\alpha_r}$—the coefficients that capture differences in purchase behavior (at the game and feature level) that are not absorbed by expected usage. Accordingly, the results in Section 6 should be interpreted as the effect of information design conditional on the existing levels of ownership utility, external advertising, Steam's recommendation algorithm, post-purchase learning, and any usage utility beyond one week.

[23] We infer returns by observations where a game, which was in a user's library at a certain date, disappears from the library later. In our estimation, we exclude the purchase and usage of the returned games.

4.2 Supply: information, targeting, and equilibrium pricing

This section presents the supply model used to simulate equilibrium targeted prices, which in turn determine key market outcomes such as profits and consumer surplus under alternative designs.

Information design. Given that the platform observes the complete purchase and usage histories of *each* consumer i, we assume—and later verify—that it can accurately determine each individual's true type, $r(i) \in \{1, ..., R\}$. The platform then decides what information to disclose to sellers for targeting individual consumers. Following Section 2, we characterize a platform's information design as choosing a *partition* of the set $\{1, ..., R\}$, which we denote as CR^x.

Under full information, the platform discloses the exact type of each consumer, so $CR^{\text{full}} = \{\{1\}, \ldots, \{R\}\}$. Under no information, all consumers are pooled together, and $CR^{\text{none}} = \{\{1, \ldots, R\}\}$. Between these two extremes, the platform may choose to disclose partial information by grouping types into coarsened segments. For example, grouping types 1 and 2 together and the rest separately corresponds to $CR^x = \{\{1, 2\}, \{3, \ldots, R\}\}$. The platform enforces its chosen information set across all sellers and overrides any information sellers may otherwise possess about individual consumers. The enforcement can be implemented through labeling consumers or seller interface design, as described in Appendix A.1.

Seller beliefs. We assume that sellers are fully informed about the overall distribution of consumer types. Given the platform-designed coarsened segments, each seller then forms posterior beliefs about the type of individual consumers. Recall that ψ_r denotes the ex ante probability that a randomly selected consumer is of type r. Under a coarsened information design, CR^x, the seller who observes label $cr \in CR^x$ forms posterior belief

$$\Pr(r(i) = r | r(i) \in cr) = \frac{\psi_r}{\sum_{r' \in cr} \psi_{r'}}, \tag{13}$$

and will set one price for segment $cr \in CR^x$ given this belief. For example, under $CR^x = \{\{1, 2\}, \{3, ..., R\}\}$, for each consumer i in segment $\{1, 2\}$, the seller will expect $\Pr(r(i) = 1 | r(i) \in \{1, 2\}) = \frac{\psi_1}{\psi_1 + \psi_2}$. Under no or full information, equation (13) reduces to the prior ψ_r and full certainty, respectively.[24]

[24] To elaborate on this point, with $CR^{none} = \{\{1, ..., R\}\}$, we have $\Pr(r(i) = r | r(i) \in \{1, ..., R\}) = \Pr(r(i) = r) = \psi_r$. Alternatively, with $CR^{full} = \{\{1\}, ..., \{R\}\}$, we have $\Pr(r(i) = r | r(i) \in cr \text{ where } r \in cr) = 1$.

Targeted-pricing equilibrium. Given the platform's information design, we assume that sellers play a Nash equilibrium and set one price per coarsened segment. Specifically, for *each* coarsened segment $cr \in CR^x$, seller $f = 1, ..., F$ solves the optimization problem defined in equation (14), choosing prices $\{p_{j,cr}\}_{j \in J_f}$ for each of its products to maximize the expected profit from segment cr. For ease of exposition, we omit the time subscript t_w in the notation.

$$\max_{\{p_{j,cr}\}_{j \in J_f}} \sum_{j=1}^{J} w_{j,f} \cdot p_{j,cr} \cdot \Big(\overbrace{\sum_{r \in cr} \frac{\psi_r}{\sum_{r' \in cr} \psi_{r'}} \cdot s_{j,r}\big(p_{j,cr}, \boldsymbol{p_{-j,cr}}\big)}^{\text{expected purchase probability for coarsened segment } cr} \Big),$$

$$\text{where ownership weight } w_{j,f} = \begin{cases} 0.7, & j \in J_f, f \neq \text{Valve}, \\ 0, & j \notin J_f, f \neq \text{Valve}, \\ 1, & j \in J_f, f = \text{Valve}, \\ 0.3, & j \notin J_f, f = \text{Valve}. \end{cases} \tag{14}$$

In the above, $\boldsymbol{p_{-j,cr}}$ denotes the prices of all other products in segment cr. The share $s_{j,r}$ represents the average purchase probability of game j for consumer type r, integrated over the empirical distribution of game libraries L_i among type r consumers—that is, $s_{j,r} = \int \Pr(\text{purch}_i \mid L_i, \boldsymbol{\Theta}_r) dG(L_i)$. Additionally, we capture the revenue split between Steam and third-party sellers using $w_{j,f}$. Profits equal revenues in our context due to zero marginal costs.

We emphasize that the standard Bertrand-Nash uniform-pricing equilibrium is a special case of our broader equilibrium concept, in which sellers rely solely on the prior distribution of consumer types and set $|J|$ uniform prices across all individuals in the market (Berry et al., 1995; Nevo, 2000). In contrast, under a general information design CR^x, the sellers' optimization problem gives rise to $|J| \times |CR^x|$ first-order conditions—one for each product-segment combination:

$$\sum_{r \in cr} \frac{\psi_r}{\sum_{r' \in cr} \psi_{r'}} \cdot \Big(s_{j,r}\big(p_{j,cr}, \boldsymbol{p_{-j,cr}}\big) + p_{j,cr} \cdot \frac{\partial s_{j,r}\big(p_{j,cr}, \boldsymbol{p_{-j,cr}}\big)}{\partial p_{j,cr}} + \underbrace{\sum_{j' \neq j} \frac{w_{j',f}}{w_{j,f}} \cdot p_{j',cr} \cdot \frac{\partial s_{j',r}\big(p_{j,cr}, \boldsymbol{p_{-j,cr}}\big)}{\partial p_{j,cr}}}_{\text{cannibalization on } j'} \Big) = 0. \tag{15}$$

Using equation (15), we solve the equilibrium price vector under each possible information design CR^x, based on the R consumer types estimated following Section 4.2.[25] Following Berry et al. (1995), we assume the existence of a pure strategy equilibrium and verify this assumption

[25] In all counterfactual simulations, we solve for the equilibrium prices for each week and then average these prices across weeks. We verify that this approach produces similar results to an alternative where we first aggregate demand across weeks and then solve for one static price vector for the entire sample.

numerically. Our numerical results also indicate that the equilibrium is unique: using 100 different initial price vectors drawn from a Halton sequence, we find that all starting prices converge to the same solution.[26]

Discussions. We now discuss several assumptions on the supply side model, imposed to maintain tractability. First, we assume that all sellers know the true distribution of consumer types—that is, the primitives and the population share of each type—and use this information to calculate expected profits. This assumption is standard in the literature (e.g., Berry et al., 1995) and is plausible in our context, given the maturity of the video game market, the availability of accessible industry reports, and the experience level of participating sellers. Even small studios typically employ professionals with years of industry experience (Carter, 2024; Clement, 2024; MarketResearch, 2025).

Second, we focus on platform information designs in which information about individual consumers is kept symmetric across sellers. These symmetric designs closely resemble common industry practices, where platforms provide a standard seller interface featuring a common list of consumer segments or a standardized set of consumer labels for targeting. We assume that the use of such common information set is strictly enforced through platform technology, which also serve as a commitment device: all sellers share the belief that no one can bypass the restrictions to target individual consumers. Nevertheless, our equilibrium framework can be naturally extended to settings where sellers have and can use different information to infer consumer types, provided there is common knowledge of all competing sellers' information sets. We present this more general case in Appendix D and presents a specific example of extreme asymmetry in information between large and small sellers in Section 6.

Third, we assume that sellers maximize profits from upfront sales each week, abstracting away from in-game purchases, in-game advertising, and dynamic pricing. This assumption enables us to simulate equilibrium outcomes across an exhaustive range of information designs while accommodating a reasonably large number of sellers and products, complex market structures, and rich consumer heterogeneity across multiple dimensions. Other than being necessary, this assumption is also plausible in our context. As discussed in Section 3, few games on Steam offer in-game purchases or carry in-game advertising. Furthermore, as discussed in Section 3 and Appendix B, the extent of price decline over time-since-launch is very small, suggesting a limited extent of price skimming (Nair, 2007) in this context.

[26] We conduct this analysis for all focal equilibria discussed in Section 6, including the full-information case, the no-information case, and the optimally coarsened information designs.

Finally, one might wonder whether consumers would perceive targeted prices as unfair (Kahneman et al., 1986) and consequently churn from the market. Although we do not explicitly model sellers' decisions regarding regular pricing versus discounts, we envision that platforms implement targeted pricing through the distribution of personalized coupons or discounts, which potentially alleviates consumers' animosity when they are charged a different regular price than the others. We interpret the segment-specific prices as effective prices net of discounts, following Besanko et al. (2003); Dubé et al. (2017); Shiller (2020); Smith et al. (2023). Specifically, a seller could set the regular price at the highest targeted price level and issue discounts to the remaining consumer segments. Additionally, although our model does not explicitly incorporate consumer fairness concerns, we address this issue by showing in Section 6 that the optimal coarsened information designs we propose help alleviate potential fairness concerns by reducing the dispersion in effective prices.

5 Estimation results

We present the composition of consumer types and their heterogeneity based on our estimated latent class model with $R = 8$. We select $R = 8$ as our final specification because it provides a good fit to the observed behavior (see Appendix Table C.2) while keeping the computational burden manageable. Note that the number of information designs, and hence the computation burden in counterfactual simulations, increases super-exponentially with the number of types. For instance, increasing the number of types to $R = 9$ would increase the number of possible information designs from $4,140$ to $21,147$.[27]

We report the full set of parameter estimates for $\boldsymbol{\Theta_1}$ and $\boldsymbol{\Theta_2}$ in Appendix D. In this section, we discuss the heterogeneity across consumer types, evaluate our model fit, and examine the model-implied price elasticities.

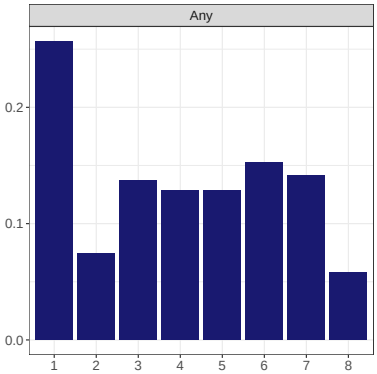
Consumer heterogeneity. We first summarize the heterogeneity in the overall engagement across latent consumer types. Figure 5.1 presents, for each consumer type (labeled 1 to 8), the model-implied average purchase probability across all games in panel (b) and the average price sensitivity in panel (c). We observe substantial heterogeneity reflected in both measures. In particular, Type 1 consumers (14% of the market) exhibit the highest level of purchase probability, buying more than one game per month on average. They are also the least price sensitive, with an average

[27] The number of possible information designs corresponds to the Bell number, which grows super-exponentially with the number of consumer types, as discussed in Appendix A.2.

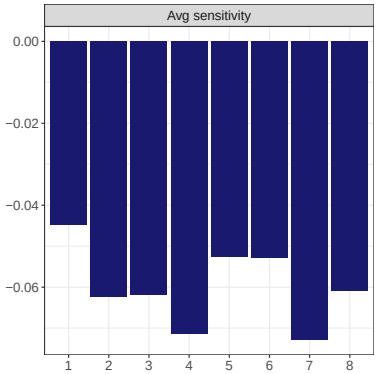
price sensitivity of −0.04. Meanwhile, Types 2 and 8 (collectively 30% of the market) purchase games far less frequently, averaging one every three months, and display moderate price sensitivities. In contrast, Types 4 and 7 (16% of the market) are the most price-sensitive types—almost twice as price sensitive as Type 1 consumers—while purchasing games half as often.

(a) Fraction of Types

Type r	Fraction ($\hat{\psi}_r$)
Type 1	14%
Type 2	11%
Type 3	12%
Type 4	8%
Type 5	22%
Type 6	6%
Type 7	8%
Type 8	19%



(b) Purchase Probability



(c) Price Sensitivity

Figure 5.1: Type distribution, purchase probability, and price sensitivity

Panel (a) shows the fraction of consumers belonging to each type; panel (b) reports the model-implied probability that each type purchases any game; and panel (c) displays the average price sensitivity across all games for each type.

We then explore consumer heterogeneity in tastes for game features. Figure 5.2 presents the fraction of purchases with the three most popular features—"fantasy," "multiplayer," and "military." We find that consumer preferences differ markedly toward game features. Moreover, there is a correlation between the horizontal and vertical dimensions of consumer heterogeneity—namely, between feature preferences and overall engagement levels. For example, Type 8 consumers, the least engaged consumer type, display a clear preference toward games with the "multiplayer" and "military" feature. Types 1 and 6, both highly engaged, disproportionately favor games with the "fantasy" feature in purchases. Meanwhile, preferences also diverge across types for other features such as "team," "adventure," "survival," and "dark," as suggested by Tables in Appendix D.2.

As discussed in Section 4.1, the model estimates capture not only inherent heterogeneity in consumer demand but also the influence of demand shifters through other channels. Although our model remains largely agnostic about the exact mechanism(s) shaping consumer demand, we further investigate the relationship between usage-stage and purchase-stage coefficients in Appendix D. The observed correlation suggests that these parameters likely reflect variation in consumers' consideration sets. Specifically, consumers are more price sensitive to games with the features they prefer to purchase and play, indicating that they pay more attention to the prices of games

with higher match values. This pattern aligns with Steam's recommendation algorithms, which tailor game displays based on users' purchase and usage histories—both reflective of persistent heterogeneity. In our counterfactual simulations of alternative information designs, we assume consumers' consideration process is table, which is plausible given that the recommendation algorithm is independent of market prices.

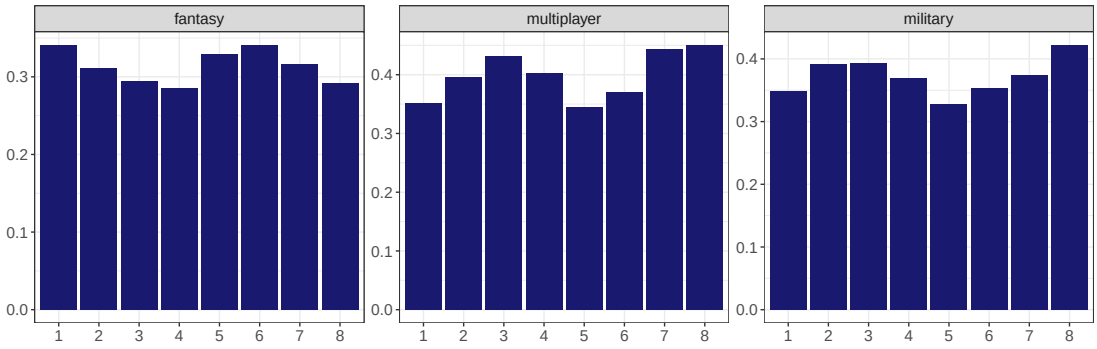


Figure 5.2: Fraction of purchases of games with popular features

Note: The panels report the model-implied probability that each consumer type purchases a game with a given feature, conditional on making a purchase. We present results for the three most popular features among all purchases.

Model fit. To examine model fit, we compare model-implied purchase behavior for each consumer type with the empirical purchase probabilities among consumers of that type. To this end, we first infer the posterior consumer types given each consumer's purchase history, using $\Pr(r(i) = r|purch_i, usage_i) = L_i(r)/\sum_{r'=1,\ldots,R} L_i(r')$ (Kamakura and Russell, 1989). We find that 95% consumers can be assigned to the most likely type with a posterior probability exceeding 0.9, suggesting that our long panel data on both purchases and usage can accurately determine consumer types. This result supports our assumption later that the platform has full information about consumer types, given that the platform likely has access to even richer data. We then label each consumer with the most likely type and contrast each type's empirical purchase behavior with the model-predicted one. As shown in Figure 5.3, our model closely matches the type-specific purchase patterns overall and across games with different features.

Further, we also show that the optimal uniform equilibrium prices match with the observed prices on Steam during our sample period. We compute equilibrium prices using our demand estimates and assuming all sellers set revenue-maximizing uniform prices according to equation (15). We demonstrate in Table 5.1 that the model-implied equilibrium prices match well with the

empirical price distribution. The average equilibrium price is $23, the median is $18, close to the observed average and median of $26 and $19. Other quantiles of the price distribution are also similar between model-implied and observed values, and so are the revenue distribution between large and small sellers and the platform. These results suggest that our demand model and the equilibrium pricing model provides a good approximation to the observed market outcomes.

Table 5.1: Fit: Observed v.s. Uniform equilibrium

Equilibrium Outcome	Data	Estimated (Uniform)
Price ($): Average	25.66	23.26
Price ($): 10 percentile	9.92	11.45
Price ($): 25 percentile	14.99	13.75
Price ($): median	19.16	17.68
Price ($): 75 percentile	34.61	26.74
Price ($): 90 percentile	52.27	50.99
Revenue Share - Valve	3.47%	3.44%
Revenue Share - Large seller	42.71%	39.45%
Revenue Share - Small seller	53.82%	57.11%

Note: We report the distribution of prices across games in the structural sample, which includes the 171 most popular titles during the sample period. For each game, we calculate the average price across all weeks it appears in the data. Within the structural sample, sellers are classified as large if they offer more than 25 games on Steam, and small if they offer 25 or fewer. We calculate the revenue shares attributed to games published by Valve, third-party large sellers, and third-party small sellers.

Implied price elasticities and substitution. How does the estimated heterogeneity translate into incentives for targeted pricing? Since a game's optimal price depends directly on its own-price elasticity—holding the prices of other games constant—we examine how these elasticities vary across consumer types. Figure 5.4 visually summarizes the price elasticities for each consumer type for each game, with darker colors representing higher elasticity in magnitude and lighter colors representing lower elasticities. The average own elasticity across games is -1.32, and the median is -1.12.[28] We observe substantial heterogeneity in elasticities across consumer types, consistent with variation in their average price sensitivities. Type 1 is the least price elastic type, with an average elasticity of -1.02. In contrast, Type 7 is the most price elastic, with an average elasticity of -1.78. Importantly, different consumer types also exhibit greater price sensitivity to different games—a pattern driven by heterogeneity in both tastes for feature and price sensitivities. For example, for

[28]These estimates indicate slightly lower price elasticity compared to values reported in the previous literature (Haviv et al., 2020), likely because our structural sample focuses on relatively popular games on Steam.

the most-elastic Type 7, the interquartile range of game-specific elasticities spans from -1.02 to -2.33.

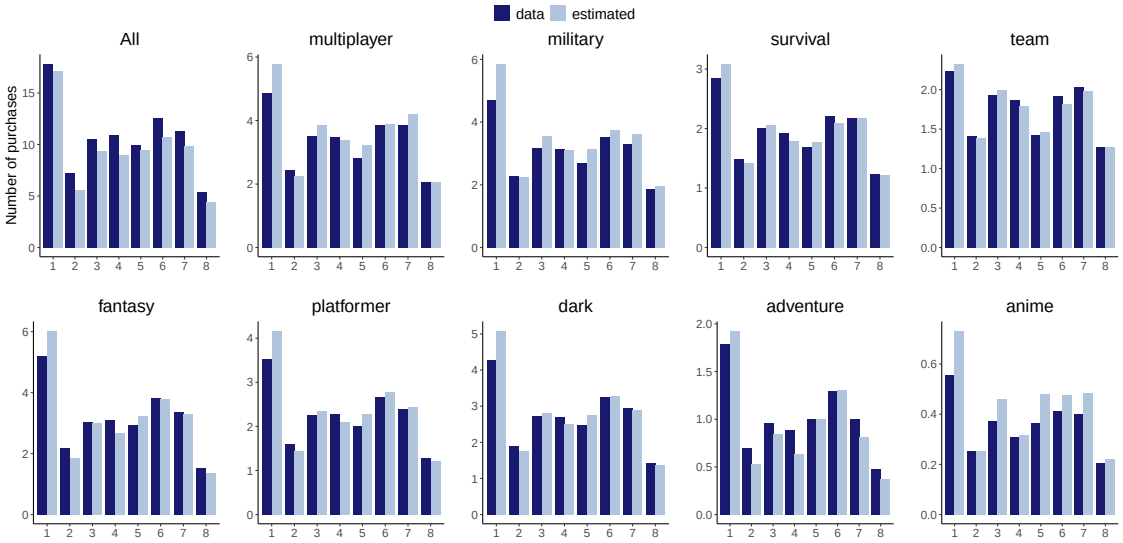


Figure 5.3: Consumer heterogeneity: purchases

Note: this figure presents moment fit regarding (1) total number of purchases, (2) number of purchases that involve certain meta-tag. The horizontal axis represents the eight consumer types.

It may appear that the elasticity matrix presented in Figure 5.4 is sufficient to inform the platform's information design for targeted pricing. After all, absent seller competition, a game's own-price elasticities across consumer types could directly inform how that game should vary prices by consumer. However, our model estimates imply that, on average, removing a game from the market results in 18% of its market share being diverted to other games. This diversion ratio varies significantly across consumer types—ranging from 7% to 28%—with Type 1 consumers exhibiting the highest diversion, suggesting that competition is intense for this highly engaged consumer type. When information design segments the market into submarkets (e.g., if it isolates Type 1 into its own submarket), it changes the intensity of competition in each submarket and among different sellers. Therefore, information design potentially has an important impact on competing sellers' pricing strategies. In the next section (Section 6), we demonstrate the importance of these equilibrium considerations.

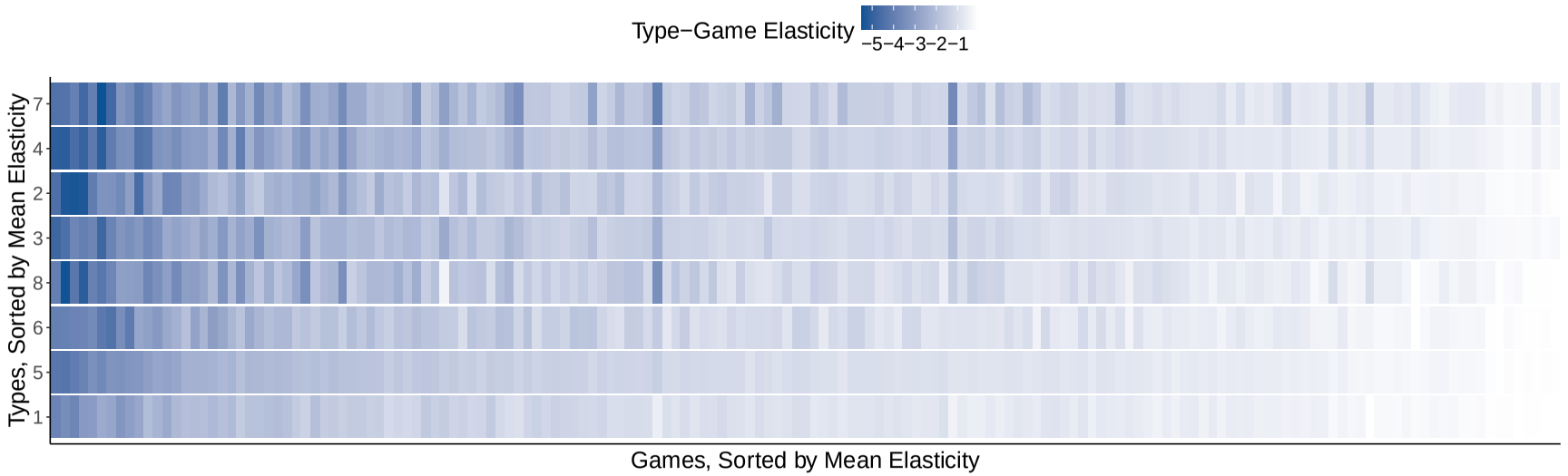


Figure 5.4: Own elasticities across games and consumer types.

Note: We evaluate all price elasticities at the observed prices. The darker a tile is, the more price-elastic the type is towards the game. We sort the types by their mean elasticity (across games) and sort the games by their market-level elasticities.

6 Designing Sellers' Information for Price Targeting

Combining the demand estimates from Section 5 with the equilibrium pricing model in Section 4.2, we exhaustively simulate counterfactual equilibrium outcomes across all possible information designs. We begin by evaluating how information design impacts profits at varying levels of granularity of segmentation (Section 6.1). We focus on the combined profits of itself and all third-party sellers, because third-party sellers account for 98% of total transaction value (Section 3), and the platform charges 30% of commission fees over this value, essentially aligning the platform's profit with sellers' total profits. Next, we demonstrate that segmenting types using demand-based heuristics alone fails to approximate the profit increases achieved by the optimal, profit-maximizing, information design (Section 6.2). We then demonstrate the robustness of our key finding—the necessity of accounting for seller competition—across alternative distributions of consumer heterogeneity (Section 6.3) and under objectives different from total profits (Section 6.4).

6.1 The profit impact of platform information design

We begin by evaluating all $4,140$ possible information designs, each as a distinct way of partitioning the eight consumer types into segments. In practice, a platform may prefer a limited number of coarse segments to reduce implementation or menu costs (Sheshinski and Weiss, 1977), protect consumer privacy (Goldfarb and Tucker, 2019),[29] or address fairness concerns (Kahneman et al., 1986) by restricting price dispersion. As such, we focus on comparing information designs with the same number of segments n (denoted as "Coarse n")—different information designs with a similar level of granularity. Figure 6.1 presents the distribution of the percentage increase in total market profits achieved by various "Coarse n" designs relative to the no-information benchmark (uniform pricing equilibrium). Note that "Coarse 8" design is equivalent to full information price targeting, where all sellers can target the true type of each consumer. "Coarse 2" to "Coarse 7" designs involve the platform grouping some consumers into coarsened segments, allowing sellers to set targeted prices based on partial consumer information.

We highlight two observations in Figure 6.1. First, price targeting under any information design yields positive incremental profits relative to uniform pricing. The highest profits arise from full-information targeting—the most granular design—which increases total seller profits by 2.88%.

[29] For instance, as described in Appendix A.1, Google's switching from FLoC to the Topics API reduces the granularity of consumer information by replacing fine-grained behavioral cohorts with broader, rotating interest categories, enhancing privacy while preserving targeting utility.

The profit increase translates to approximately $123.8 million in additional annual profits based on Steam's estimated $4.3 billion total revenue in 2017.[30] This substantial profit increase suggests that targeted prices' ability to capture consumer heterogeneity outweighs their impact on price competition, alleviating concerns about intensified competition when sellers target (see Holmes 1989; Corts 1998; Dubé et al. 2017, and our conceptual example in Section 2).

Second, whereas targeting under any information design improves profits directionally, the magnitude of profit increase varies substantially across specific designs, even when the granularity, n is held constant. As illustrated by the box plots in Figure 6.1, the median percentage profit increase across all possible "Coarse 2" designs is only 0.45%, with the minimum being 0.05%, both substantially below the 1.74% achieved by the optimal "Coarse 2" design. Likewise, "Coarse 3" and "Coarse 4" designs' median profit increases are 0.96% and 1.41%, respectively, with minimum increases of just 0.17% and 0.39%—much lower than the optimal "Coarse 2" design. These patterns underscore that platform information design is not merely about the granularity of segmentation. Rather, *how* to segment given the granularity can substantially affect the incremental profit from targeting.

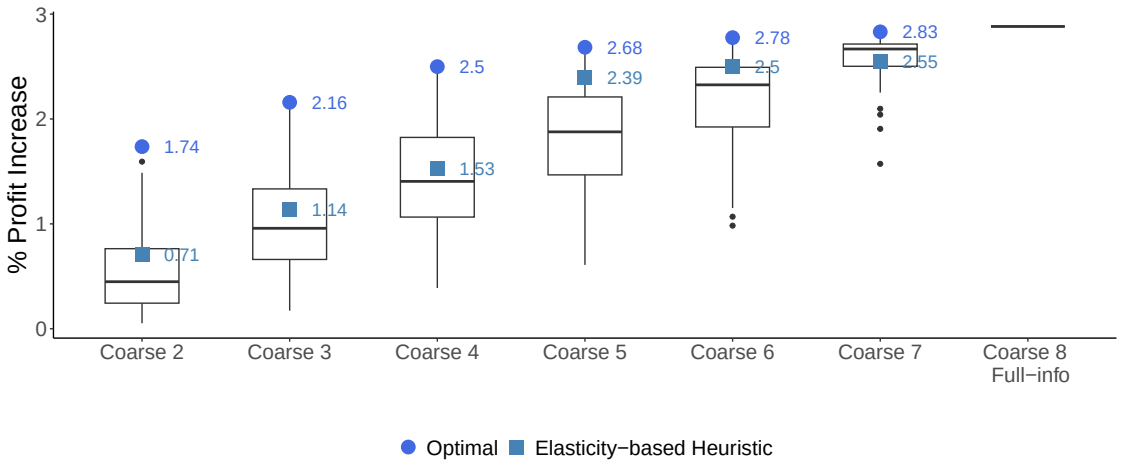


Figure 6.1: Profit increases from coarsened information designs

Note: The box plots demonstrate how the profit increases from targeting vary across alternative information designs given the same number of segments, n. The lines in the boxes indicate medians, while the bins indicate the inter-quartile ranges. The blue circle indicates the optimal design achieved by our comprehensive search across all possible designs, whereas the steel blue square indicates the profit level achieved by the elasticity-based heuristic approach.

[30] Sergey Galyonkin, who runs a third-party monitoring website SteamSpy, estimated Steam's total sales revenue at $4.3 billion for 2017. Source: `https://mcvuk.com/business-news/steam-made-4-3b-in-game-sales-in-2017-50-from-top-100-games`.

6.2 Understanding the importance of equilibrium considerations

We have shown that, given the number of segments n, the optimal (total-profit-maximizing) design, identified through counterfactual simulations that explicitly account for each information design's impact on equilibrium pricing strategies, generates substantially higher profits than most alternative designs. This finding leads to a natural question: how important is equilibrium consideration in information design? Can practitioners use demand heuristics to achieve comparable performance?

To answer this question, we use a heuristic approach that clusters consumer types into coarse segments solely on the similarity of their own-price elasticities across video games. Specifically, for each n, we calculate pairwise Euclidean distances between the type-specific elasticity vectors and then apply k-means clustering to minimize the within-cluster sum of squared distances. Because own-price elasticity is closely related to optimal single-product pricing, this procedure ensures that types that are grouped into the same segment are similar in the demand metric that is relevant to most games' pricing.[31] We present the profit increases achieved by this elasticity-based heuristic using the steelblue squares in Figure 6.1.

We find that elasticity-based heuristic designs consistently underperform the optimal information designs for each level of granularity n. The profit gap is especially pronounced when n is small—cases in which the platform deliberately limits seller information through coarsened segments with low granularity. For "Coarse 2" to "Coarse 4" designs, the gap in percentage profit increase is approximately one percentage point, corresponding to annual profit differences of \$42–44 million. As granularity increases, the gap narrows but remains substantial; even with "Coarse 7," the profit gap is 0.17 percentage points, or roughly \$7.3 million annually.

We then present what segments are formed under the elasticity-based heuristic approach and our optimal information designs in Table 6.1. The differences between the two approaches are evident. Elasticity-based heuristics tend to isolate consumer type(s) that exhibit distinct elasticity patterns from the rest of the consumer types. By contrast, the optimal designs sometimes pool these distinct types with others. For instance, the heuristic "Coarse 2" design separates Type 7—the most price-elastic group for most games—from the other types. In contrast, the optimal design combines Type 7 with four other types that are closer in the average price elasticity level but differ in their preferences for specific game features. This result deviates from the intuition from monopolistic pricing settings, where recent literature emphasizes isolating distinct consumer

[31] Visually, this clustering approach is akin to finding similarly colored *rows* in Figure 5.4

types using increasingly rich demand-side variables or advanced models to estimate heterogeneous preferences or treatment effects precisely (Shiller, 2020; Smith et al., 2023; Zhang and Misra, 2022). With competing sellers, however, optimal information design for price targeting must account for the nuanced impact of segmentation on how sellers compete "locally" within each segment, and hence their competitive pricing strategies.

Table 6.1: Profitability Comparison: Optimal v.s. Heuristic

Heuristic Segmentation	% Profit gain	Optimal Segmentation	% Profit gain
$\{1,2,3,4,5,6,8\},\{7\}$	0.71	$\{1,5,6\},\{2,3,4,7,8\}$	1.74
$\{1,2,3,5,6,8\},\{4\},\{7\}$	1.14	$\{1\},\{2,5,6,8\},\{3,4,7\}$	2.16
$\{1,2,3,5,6\},\{4\},\{7\},\{8\}$	1.53	$\{1\},\{2,8\},\{3,4,7\},\{5,6\}$	2.50
$\{1\},\{2,3,5,6\},\{4\},\{7\},\{8\}$	2.39	$\{1\},\{2,8\},\{3\},\{5,6\},\{4,7\}$	2.68
$\{1\},\{2,3,5\},\{4\},\{6\},\{7\},\{8\}$	2.50	$\{1\},\{2\},\{3\},\{5,6\},\{4,7\},\{8\}$	2.78
$\{1,5\},\{2\},\{3\},\{4\},\{6\},\{7\},\{8\}$	2.55	$\{1\},\{2\},\{3\},\{4\},\{5,6\},\{7\},\{8\}$	2.83

Note: we measure the profitability of a given segmentation by its % incremental profit increases relative to the uniform pricing equilibrium. For readability, in this figure, we suppress the outer braces and braces for singleton sets. For example, we denote information set $\{\{1,2,3,4,5,6,8\},\{7\}\}$ by $\{1,2,3,4,5,6,8\},\{7\}$.

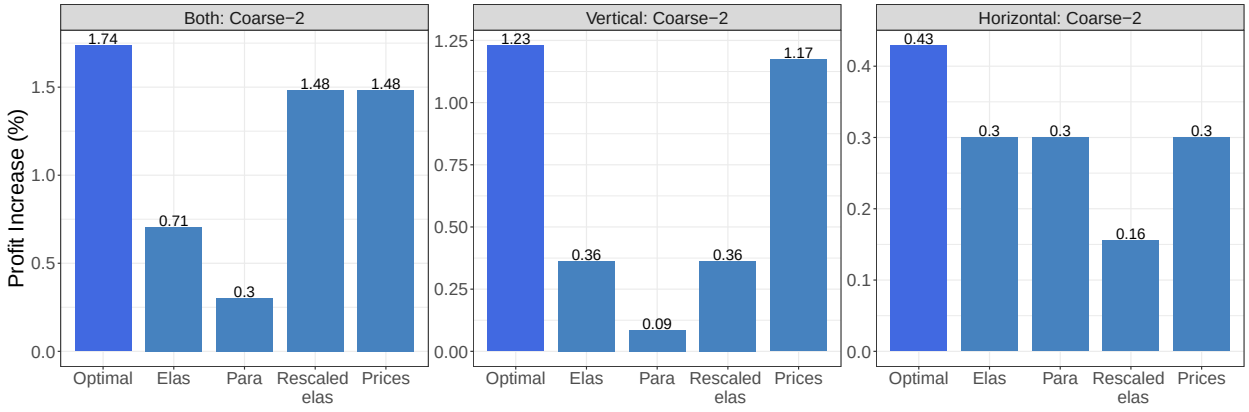


Figure 6.2: Profit gaps: alternative heuristics and alternative demand.

Note: This figure compares the profit increases achieved by various heuristic approach under alternative distributions of consumer heterogeneity with $n = 2$. We use three different heuristics to cluster consumers into coarsened segments: their own-price elasticities across games ("Elas"), the full vector of utility parameters ("Para"), and the rescaled own-price elasticities relative to each game's most elastic type ("Rescaled elas"). We also include a heuristic based on full-information equilibrium prices ("Prices"). The left panel shows results under the estimated (factual) demand distribution with both vertical and horizontal heterogeneity; the middle panel uses a counterfactual distribution with isolated vertical heterogeneity; and the right panel uses a distribution with isolated horizontal heterogeneity. In this and subsequent figures, we use blue bars to highlight the "Optimal" design and steel blue bars for all other approaches or cases.

One might question whether own-price elasticities across games are the most appropriate metric for capturing demand heterogeneity across consumer types. To address this, we further construct two alternative demand-based heuristics. First, we cluster consumer types based on their type-specific demand parameters ($\boldsymbol{\beta_r}$, $\boldsymbol{\alpha_r}$, and $\boldsymbol{\gamma_r}$). Second, we cluster based on rescaled own-price elasticities, which adjust for the differences in price elasticities across games.[32] The left panel of Figure 6.2 shows that neither of the two alternative measures ("para" and "rescaled elas") aligns with the optimal information design and, as we demonstrate later, neither consistently approximates the optimal design across alternative demand structures. These results further underscore that segmenting consumers based solely on similarities in preferences or price elasticities is insufficient for effective information design.

One might further conjecture that heuristics incorporating some supply-side factors—such as the distribution of competing sellers and the positioning of differentiated products in the product feature space—could help approximate the optimal design without simulating all counterfactual information designs. A natural heuristic that reflects these considerations is to cluster consumers based on their type-specific equilibrium prices under full information. These prices capture both demand heterogeneity and equilibrium pricing incentives under the most granular information design. However, as shown in the left panel of Figure 6.2, this approach still underperforms the optimal information design. We conjecture that this gap persists because coarsened information designs induce equilibrium price responses that are specific to the coarse market partitions they create. Hence, they cannot be inferred from the equilibrium prices under full information, which only captures the competition under a specific information design.

6.3 Unpacking consumer heterogeneity

How do different components of consumer heterogeneity contribute to the profit gap between the heuristic approach and the optimal information design? Is there a specific dimension of consumer heterogeneity that drives the previous findings? To answer these questions, we examine counterfactual equilibrium outcomes under optimal information designs and heuristics-based designs when consumer heterogeneity is limited to a single dimension.

Guided by two different theoretical frameworks, we decompose consumer heterogeneity into

[32] Specifically, for each game j, we rescale the own-price elasticities by dividing each type's elasticity by the elasticity of the most price-sensitive type for that game: $\tilde{\eta}jj^r = -\eta jj^r / \max_r |\eta^r_{jj}|$. Visually, this clustering approach is similar to finding similarly colored *rows* in Figure D.3. The darker tiles indicate higher relative elasticity—i.e., greater price sensitivity compared to other types—for a given game.

two dimensions: vertical and horizontal heterogeneity. To isolate the vertical dimension, we retain only the differences in average engagement and price sensitivity across types while removing all feature-specific taste shifters. This consumer composition mirrors the set-up in Holmes (1989), where consumers differ only vertically and all sellers compete for the same "strong" versus "weak" markets. To isolate the horizontal dimension, we equalize average engagement and price sensitivity across types but preserve variation in taste for specific game features. This consumer composition mirrors the set-up in Thisse and Vives (1988) and Corts (1998), where sellers have asymmetric incentives to target consumer types—that is, one game's "strong" market may be another game's "weak" market. In Appendix Figure D.4, we visualize the elasticity patterns corresponding to the two counterfactual consumer compositions.

Despite that the eight consumer types differ along both vertical and horizontal dimensions, we find that vertical heterogeneity is the primary driver of profit increases in Steam's market. As shown in Figure 6.3, targeting based on full information about vertical heterogeneity yields a 1.73% increase in total market profits relative to uniform pricing. In contrast, targeting based solely on horizontal preference heterogeneity leads to a more modest 0.96% increase. Nevertheless, the positive profit impact under purely horizontal heterogeneity alleviates concerns that asymmetric rankings of "strong" versus "weak" markets across sellers will lead to profit losses under targeted pricing (Corts, 1998; Dubé et al., 2017).

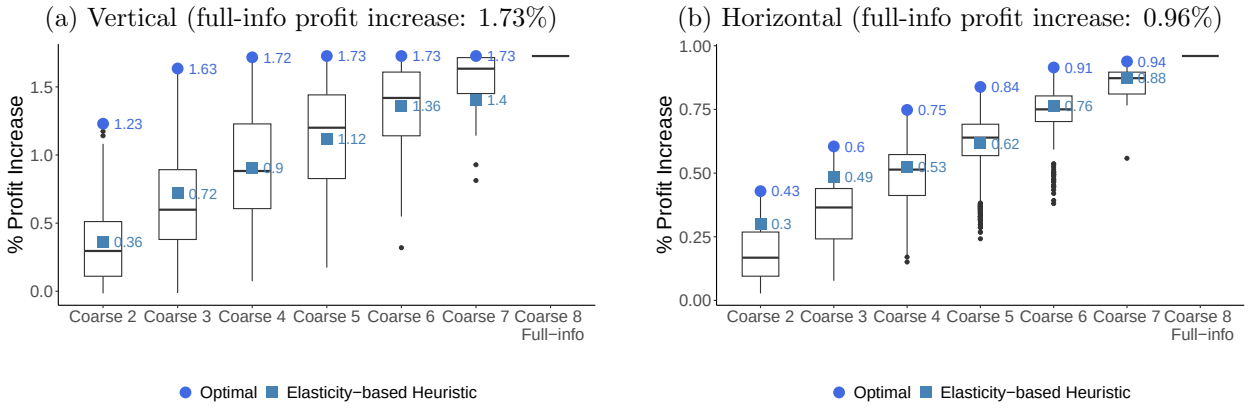


Figure 6.3: Unpacking heterogeneity: profit gaps persist

Note: The box plots demonstrate how the profit increases from targeting vary across alternative information designs given the same number of segments, n, under the two counterfactual consumer compositions. As in Figure 6.1, the blue circle indicates the optimal design achieved by our comprehensive search across all possible designs, whereas the steel blue square indicates the profit level achieved by the elasticity-based heuristic approach.

More importantly, our key finding—that clustering consumers on similarity in demand alone

fails to approximate the optimal information design—is robust across both counterfactual consumer compositions. As shown in Appendix Figure D.4, even with pure vertical heterogeneity, elasticity-based heuristics only capture 29% of the potential profit increases under the optimal "Coarse 2" design (0.36% vs. 1.23%). This result suggests that even with simple lower-dimensional heterogeneity and associated elasticity patterns, and with directionally aligned seller pricing incentives, demand-based heuristics still underperforms the optimal design.

We further explore the performance of these heuristics using alternative demand-based metrics and full-information equilibrium prices in the right two panels of Figure 6.2. Among the three demand-based heuristics, none consistently approximates the performance of the optimal information design. This suggests that the gap is not due to the complexity of demand alone. However, we do observe that segmentation using full-information equilibrium prices is closer to the optimal information design, especially when consumer heterogeneity is purely vertical. Taken together, these findings underscore the critical importance of incorporating supply-side considerations when designing information for competing sellers, even when consumer heterogeneity is relatively simple and limited to a single dimension.

6.4 Other platform objectives

Our earlier analyses in Sections 6.1 through 6.3 have focused on total profits as the primary platform objective. However, it is important to note that our equilibrium framework can also be used to evaluate alternative platform objectives that depend on equilibrium prices. In this section, we examine how well the optimal information design for $n = 2$—selected by maximizing total platform and seller profits—aligns with other objectives the platform may value. Specifically, we consider three alternative objectives that reflect long-term platform goals or potential incentive misalignments: (1) consumer welfare, (2) profit redistribution between large and small sellers, and (3) the platform's own profit, which includes 100% of profits from Valve's own games and 30% of profits from third-party titles.

Consumer welfare. The platform may care about consumer welfare, which could affect user retention and, in turn, platform growth in the long run. We measure consumer welfare as the inclusive value of owning video games and compare the changes in welfare under targeting relative to the uniform pricing equilibrium.[33] Figure 6.4 presents these changes across three information

[33] We do acknowledge that our consumer welfare calculation assumes the estimated demand parameters reflect true underlying preferences, an assumption not required for evaluating counterfactual prices and profits under different

designs: the elasticity-based heuristic "Coarse 2" design, the optimal "Coarse 2" design, and the full information design. We find that full information design increases consumer welfare by 115 utils, representing a 1.72% improvement over the uniform pricing baseline of 6,685 utils. Notably, the optimal "Coarse 2" design captures most of the welfare increase despite its reduced granularity. In contrast, the heuristic "Coarse 2" design yields little to no improvement in consumer welfare relative to the baseline, despite its same level of granularity as the optimal "Coarse 2" design.

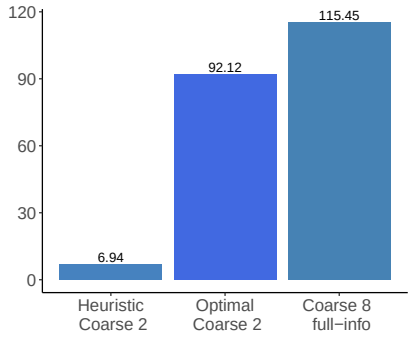


Figure 6.4: Consumer Welfare Change Relative to the Uniform Pricing Benchmark (in Utils)

What makes the profit-maximizing optimal "Coarse 2" design also effective in improving consumer welfare? In Appendix Table D.7, we further examine how each information design impacts each consumer type. We find that the optimal "Coarse 2" design significantly lowers the average price charged to segment $\{2, 3, 4, 7, 8\}$ (58% of all consumers) while only bringing a modest price increase for segment $\{1, 5, 6\}$ (42% of all consumers). Specifically, the former group receives an average price discount of 21.8% relative to the baseline uniform price—approximately \$6 per game—resulting in substantial welfare gains. In contrast, the latter group sees only a 5.5% price increase—about \$0.89 per game—yielding limited welfare loss. In other words, the optimal "Coarse 2" design causes prices to fall sharply for the majority of types, thereby increasing total quantity and consumer welfare. This finding is in line with Dubé and Misra (2023)'s results from a monopoly market. It is worth noting that our measure of consumer welfare does not incorporate intrinsic preferences for privacy (Lin, 2022) or concerns about price fairness. If these factors were accounted for, the "Coarse 2" design would likely further approximate—or even surpass—the consumer welfare achieved by the full-information design, while maintaining a clear advantage over the heuristic approach.

information designs.

Information symmetry between large versus small sellers. Recent discussions have raised concerns that larger sellers—owing to their broader product portfolios and longer operating histories—may possess an informational advantage over smaller sellers. If exploited, this advantage could allow large sellers to crowd out smaller competitors, ultimately diminishing product variety on the platform over time.[34]

In light of this discussion, we examine a concerning asymmetric information structure, where large sellers (those with more than 25 games) hold and can use full information for targeting, but the small sellers (25 games or fewer) hold no information and cannot target at all. This represents a scenario where the platform does not actively design information or imposed common segmentation for sellers, but instead allows sellers to use their own information about consumers. We compare this asymmetric targeting case to three key information designs—heuristic "Coarse 2," optimal "Coarse 2," and full information. As we show in Figure 6.5, this asymmetric information scenario would significantly benefit large sellers at the expense of small sellers. In contrast, the optimal "Coarse 2" design achieves a more balanced distribution of profit increases across seller sizes, while also delivering a comparable total profit to the asymmetric case. Therefore, the symmetric, profit-maximizing "Coarse 2" design also aligns with the platform's incentive to level the playing field between large and small sellers.

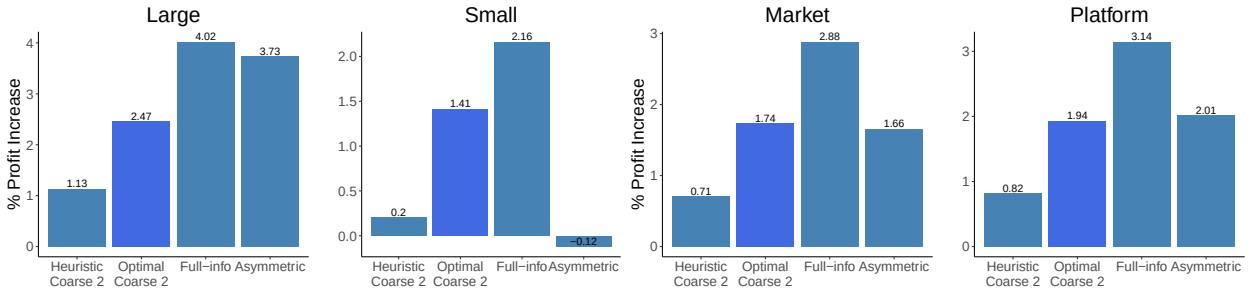


Figure 6.5: Alternative objectives: profit increases for different parties.

Note: This figure presents profit increases under alternative information designs for different stakeholders: third-party large sellers, third-party small sellers, the entire market (platform plus third-party sellers), and the platform itself. We use the uniform-pricing equilibrium as the benchmark.

Platform self-preferencing. Further, we examine the impact of different information designs on platform profits versus third-party seller profits. As we have discussed in Section 3, Valve

[34] A growing body of literature suggests that the asymmetric effects of policies like Europe's General Data Protection Regulation (GDPR) could provide large firms with an extra competitive edge over the smaller firms (Campbell et al., 2015; Bourreau et al., 2017; Jia et al., 2018; Bleier et al., 2020; Johnson et al., 2023; Zhao et al., 2021), exacerbating the disparities between large and small sellers.

serves both as a platform, which collects commission fees from third-party sales, and a seller, which develops and sells its own games. Recent literature and policy discussions are concerned about platform designs that serve the platform's own interest rather than sellers' interests (Congress, 2021; European Commission, 2022). We examine whether Valve's dual role creates incentives to adopt an information design that maximizes its own profits at the expense of the broader marketplace. We find that the profit-maximizing information designs for Valve given each n are identical to those reported in Table 6.1, which maximize total profits. Moreover, as shown in Figure 6.5, the relative profit increases for Valve under the three designs follow a pattern similar to the increase in total profits. This incentive *alignment* reflects the fact that Valve's own titles constitute only a small revenue share of the entire market, as mentioned in Section 3.

7 Conclusion

We empirically study a platform's problem in designing targetable consumer segments for competing sellers, which we refer to as "platform information design." Using novel micro data from Steam, we estimate a flexible demand model that captures heterogeneous consumer demand and develop a targeted pricing model to simulate equilibrium outcomes under all counterfactual information designs with discrete consumer types. Because targetable information partitions the market and changes how sellers compete with each other, accounting for the equilibrium impact is paramount in information design. To this point, our results demonstrate that the profit-maximizing information design—which distinguishes heterogeneous consumer types while accounting for its equilibrium impact—yields substantially higher profits compared to demand-based heuristics that ignore this equilibrium impact.

Our findings offer important managerial implications for platforms that provide consumer information to competing sellers, showing that the design of targetable segments is not merely an exercise that tries to distinguish heterogeneous consumer types. Instead, the platform must also account for how its information design affects seller competition in equilibrium, especially when it intends to provide only a limited number of consumer segments to sellers.

A limitation of our study is that we use counterfactual simulations to evaluate alternative information designs, assuming that sellers set optimal targeted prices given an information structure. Our approach is necessary because it is challenging for any platform to experimentally vary its information design and let sellers adjust to a new equilibrium. Nevertheless, evaluating platform

information design using field or natural experiments would constitute a welcome topic of future research.

We also have imposed various assumptions to simplify our demand model—including limited learning about quality or match value, full consideration sets (even though we approximate limited consideration with flexible demand parameters), and a lack of forward-looking purchase behavior beyond a fixed horizon. Whereas we present evidence in support of these assumptions in our setting, migrating our framework to a different context would require one to re-examine these assumptions. For example, in an evolving market with less experienced consumers, learning or other forms of demand dynamics may be prevalent, necessitating a different demand model and potentially altering sellers' pricing incentives.

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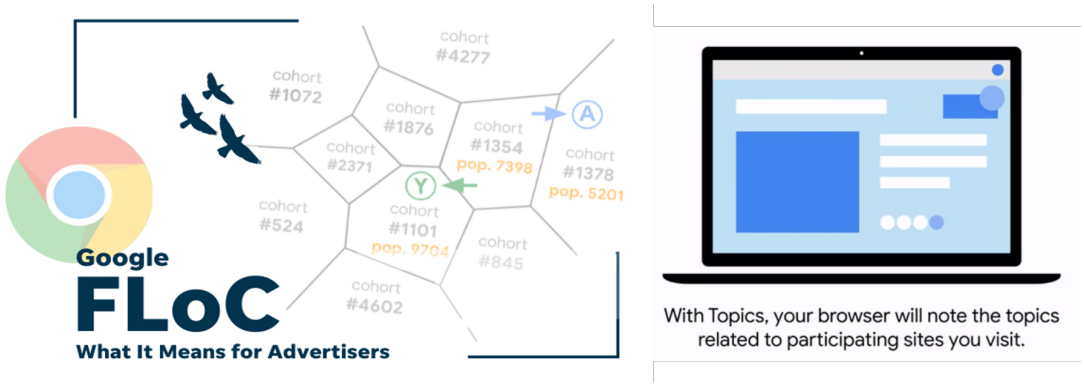
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Online Appendix

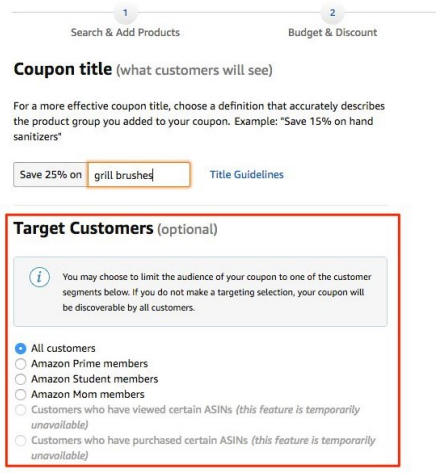
A Platform information provision in practices

A.1 Segmentation: reveal partial information about consumers for targeting

(a) Google: topics of interest



(b) Amazon: membership or history



(c) eBay: pre-defined consumer groups

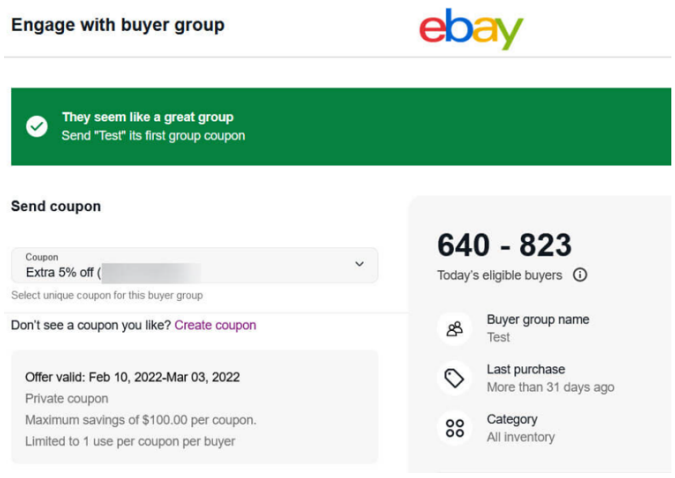


Figure A.1: Segmentation for Targeting: Google, Amazon, and eBay

Figure A.1 illustrates how platforms enable sellers (or advertisers) to target consumers in groups by providing predefined labels or segment lists. These examples suggest that information provision through segmentation is technically feasible through the platform's design of infrastructure, without transfer of raw consumer data across business entities.

A central challenge for platforms is striking the right balance between segment granularity and privacy protection. In 2022, for example, Google replaced its fine-grained Federated Learning of Cohorts (FLoC) ini-

tiative with the broader Topics API to reduce re-identification risks and enhance user privacy, demonstrating the importance of using coarser segmentation for effective targeting.

A.2 Number of possible information designs: Derivation

Let X be a finite set consisting of n consumer types in the market. A *partition* of X can be defined as a collection of disjoint nonempty type subsets of X, whose union is X. For example, for $X = \{a, b, c\}$, there are 5 possible partitions:

$$\{\{a\},\{b\},\{c\}\}, \quad \{\{a,b\},\{c\}\}, \quad \{\{a,c\},\{b\}\}, \quad \{\{b,c\},\{a\}\}, \quad \{\{a,b,c\}\}.$$

Conceptually, each partition corresponds to a particular information design which constrains how sellers can target consumer type a, b, and c. Next, we show that the number of possible partitions, and thus information designs, depends solely on the cardinality of X— n. We denote this number as $B(n)$, which is the n-th Bell number (Bell, 1934; Rota, 1964).

Proof of the recursive formula for $B(n)$. Consider a set of $n+1$ elements, and denote the $(n+1)$-th element x. Note that for any partition of the set, element x must be in one of the subsets. There are two possibilities: First, x is in a subset by itself. In this case, the remaining n elements must be partitioned, and there are $B(n)$ ways to do this. In the second case, x is in a subset with some other elements. To count the possible partitions in this case, we can partition the n elements into k subsets ($k \in \{1, 2, ...n\}$), and add x to one of these k subsets. Therefore, we have:

$$B(n+1) = B(n) + \sum_{k=1}^{n} k \cdot S(n,k), \tag{16}$$

where $S(n,k)$ indicate the Stirling numbers of the second kind, which calculates the number of ways to partition n elements into k subsets Through algebraic manipulation and the relationship between Stirling numbers and binomial coefficients, the formula of $B(n+1)$ can be transformed into:

$$B(n+1) = \sum_{k=0}^{n} \binom{n}{k} B(k), \text{ where } B(0) = 1. \tag{17}$$

Growth Rate of Bell Numbers. It is noteworthy that $B(n)$ grow faster than exponentially as n increases. To illustrate, we have the following Bell numbers following the recurrence formula:

$$\begin{aligned} B(4) &= 15 \\ B(5) &= 52 \\ B(6) &= 203 \\ B(7) &= 877 \\ B(8) &= 4140 \\ B(9) &= 21147 \\ B(10) &= 115975 \end{aligned} \tag{18}$$

The growth rate highlights the high-dimensional nature of the information design problem and the rising computational cost as the number of consumer types increases.

B Additional industry details and descriptive patterns

B.1 Competition outside Steam: negligible

One may wonder whether Steam competes against other major platforms. To check the relative size of Steam to its competitors, we collect search data from Google Trends to compare the number of search queries for Steam and its major competitors, Origin (developed by Electronic Arts), Epic Store (developed by Epic Games), and Uplay (developed by Ubisoft). As shown in Figure B.1, the search volume for all three competitor platforms are negligible compared to that for Steam, confirming Steam's near-monopoly position in the PC-gaming market.

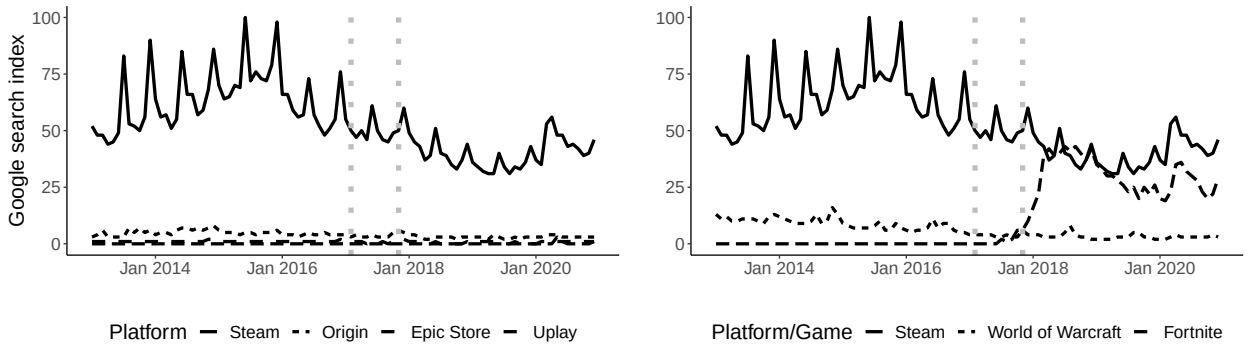


Figure B.1: Steam takes a near-monopoly position for PC-based gaming.

Note: In both figures, the gray dotted lines denote the start and end of our sample period. The vertical axis measures the Google search index, normalized by the highest value among the platforms (or games) in comparison.

We also compare the search for Steam against two major PC games outside of any platform. The first one is *World of Warcraft*, released by Blizzard Entertainment in 2004, and the second one is *Fortnite*, released by Epic Games in 2017. Both games had low search volumes during our sample period, as the craze for *World of Warcraft* had already faded, while the spike of interest in *Fortnite* had not arrived. Hence, we do not consider them as major substitutes for games inside Steam.

B.2 Tags and meta-tags

We present an example of user-generated tags in Figure B.2 and the defined Meta-tags in Table B.1.

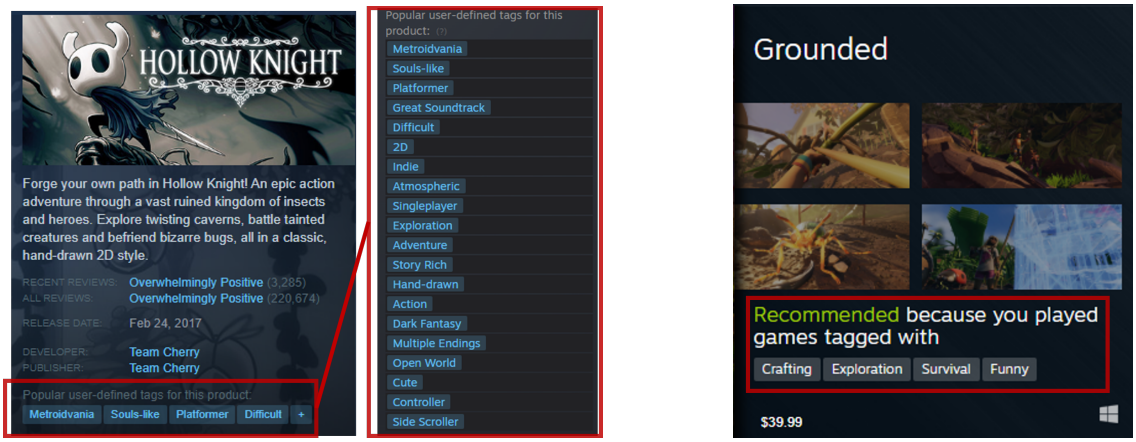


Figure B.2: Example of user-generated tags

Table B.1: Meta-tags Defined

	Meta-Tag	Number of Original Tags	Popular Original Tags
1	fantasy	25	classic, comedy, fantasy, horror, sci-fi
2	multiplayer	5	arcade, co-op, fps, mmorpg, moba
3	military	12	rts, slash, stealth, tactical, war
4	dark	20	dark, gore, nudity, sexual, violent
5	survival	17	building, crafting, management, resource, survival
6	platformer	3	isometric, platformer, tps
7	anime	7	anime, cartoon, cartoony, hentai, memes
8	team	8	city, football, sports, team, trading
9	adventure	11	agriculture, exploration, flight, pirates, space

Note: For each meta-tag, we present the top five most frequently used original tags.

B.3 Consumer Learning: negligible

One might wonder whether consumer learning plays a significant role in the video game market, especially given the nature of content goods. If consumers engage in learning— either about product quality or how well different content features match their preferences— our static demand model could misrepresent the true distribution of consumer heterogeneity. As discussed in Section 4.1, we argue that learning is minimal

in the Steam market due to the abundance of information available about game quality and the maturity of most Steam users. We further validate this claim by examining consumers' return behavior and their evolving preferences for content features over time below.

Steam offers a generous return policy, allowing consumers to return a game within two weeks of purchase (or within two days during game-specific free weekends). As we exclude all returned games from our purchase panel, we find regular returns account for only 3.8% of initial purchases, suggesting that consumers face limited uncertainty about game quality. We then analyze whether consumers change their preferences for game features over time. Specifically, for each consumer, we compare the features of their first five purchases in the sample period to those of their last five. As shown in Figure B.3, the distributions are largely similar across consumers, indicating minimal shifts in preferences and suggesting that consumer demand remains stable during our sample period.

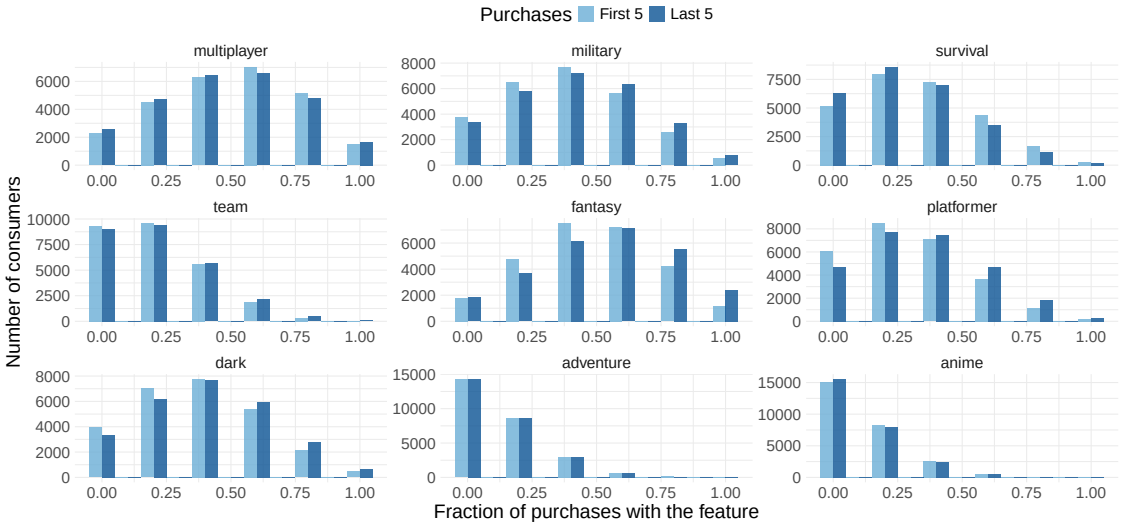


Figure B.3: What features consumers buy (first 5 vs. last 5 purchases)

B.4 Pricing patterns on Steam

We examine how typical game prices evolve over time on Steam. To capture the full pricing trajectory, we focus on the 1,434 games released between January 2015 and December 2018, tracking their price paths through December 2020 to ensure at least 24 months of observation for each title.

As shown in Figure B.4, a game's regular (list) price tends to remain stable over time. In contrast, developers frequently adjust the effective prices through discounts, with the depth of discount gradually increasing as the game ages. However, unlike the patterns observed in Nair (2007), the overall decline in effective prices is modest, with little additional change once a game is more than a year old.

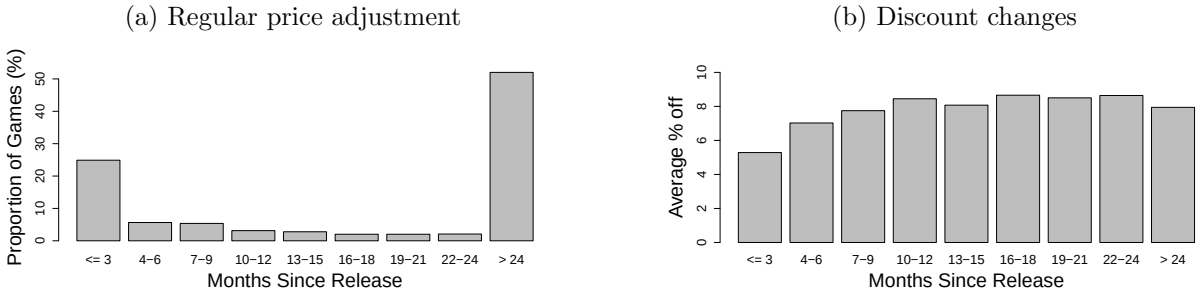


Figure B.4: Gradual price decline over time.

Price declines slowly over time. Notably, intertemporal price variation is driven primarily by discounts rather than changes in regular prices. As shown in Figure B.4a, only 25% of games adjust their regular prices within the first three months after release, while more than half never change their regular prices within the first two years. Meanwhile, although the average discount increases over time, as shown in Figure B.4b, the decline is gradual and modest in magnitude.

To further isolate within-game price changes over time, we estimate a regression of game prices on months since release, controlling for game fixed effects. Table B.2 reports results for three measures: average monthly price, lowest (effective) monthly price, and maximum monthly discount relative to the release price. All three measures show a gradual decline post-release. Average and effective prices fall by about \$0.07 per month, with discounts increasing by 0.46% per month. These modest declines suggest that consumers have limited incentives to delay purchases on Steam, in contrast to the more pronounced intertemporal pricing effects observed in the console market (Nair, 2007).

Table B.2: (Effective) Game prices decline slowly after releases.

	Average price (\$)	Effective price (\$)	Effective discount (% off)
	(1)	(2)	(3)
Month(s) Since Release	-0.07^{***}	-0.07^{***}	0.46^{***}
	(0.001)	(0.001)	(0.005)
Game FE	Yes	Yes	Yes
Observations	137,041	137,041	137,041
R^2	0.90	0.78	0.47
Adjusted R^2	0.90	0.77	0.46
Residual Std. Error (df = 135609)	4.33	5.91	27.90
Note:			*p<0.1; **p<0.05; ***p<0.01

Identifying variations for price sensitivities. To illustrate the variation that identifies price sensitivities, Figure B.5 shows the pricing path and corresponding demand for *Hearts of Iron IV* as an example. As seen in the figure, sharp price discounts are typically followed by immediate spikes in purchases.

These purchase surges around discount events help identify consumer price sensitivity. One potential concern

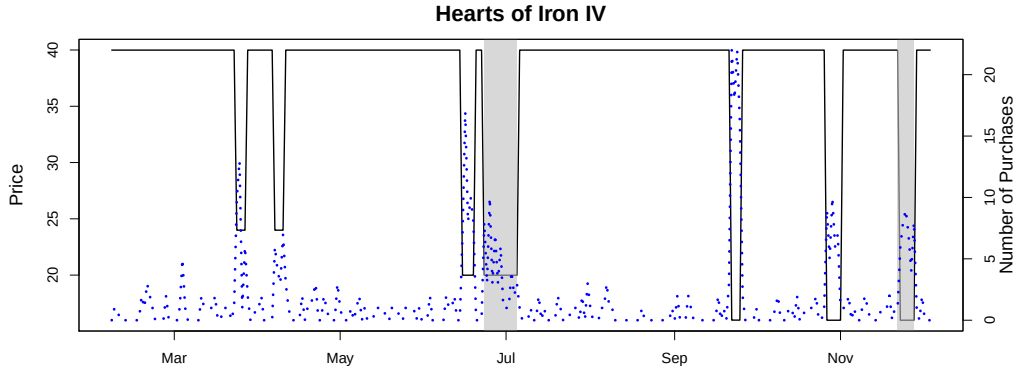


Figure B.5: Price and purchases: February 6th, 2017 to December 4th, 2017

Note: We plot the daily number of purchases among sample users against the price changes over time. The black line represents the price path, while the blue dotted lines represent the purchase path. The shaded areas represent the two major sales events on Steam — the summer sales event and the autumn sales event.

is that our estimated price sensitivities around steep discounts may reflect inventory stockpiling-driven demand spikes, thereby biasing the coefficients downward. To address this, we test robustness by controlling for major Steam events (e.g., Summer and Autumn sales).

Table B.3 reports a reduced-form specification on the same game sample used in our structural estimation. Adding indicators for the pre-sale and during-sale weeks yields only a mild difference in the price coefficient, and the pre-sale coefficient is statistically insignificant— evidence against systematic stockpiling.

Table B.3: Robustness Check: stockpiling before sales event

	log(Purchase + 1)	
	(1)	(2)
Price	-0.053^{***}	-0.047^{***}
	(0.004)	(0.003)
Pre-event		-0.083
		(0.079)
Sales-event		0.252^{***}
		(0.070)
Game-Month FE	yes	yes
Clustered S.E.	Game	Game
Observations	6,891	6,891
R^2	0.711	0.716
Adjusted R^2	0.610	0.616
Residual Std. Error	0.662 (df = 5105)	0.657 (df = 5103)
Note:		*p<0.1; **p<0.05; ***p<0.01

C Additional Model Details

C.1 Monte-Carlo Simulation

This subsection validates our demand model through Monte Carlo simulations. We consider a market with 10 video games and two consumer types—heavy (30% of the market) and light (70% of the market)—who differ in purchase intensity, usage intensity, and price sensitivity (Table C.1). Consumers make purchase decisions based on expected future usage and the additional value of expanding their game library. We simulate consumer purchase and usage behavior for 6000 consumers and 240 days, approximating our structural sample. We implement estimation for $MC = 30$.

Table C.1 reports the true parameter values (first column) and compares estimator performance across four scenarios: estimating on the full simulated sample versus only the latter half, each with and without modeling the usage stage. When applied to the full data path, the estimator recovers both homogeneous and heterogeneous parameters without bias. Restricting estimation to the second half— mirroring our empirical setting of non-empty initial game libraries— introduces only minimal bias in the recovered parameters when the usage stage is modeled.

Finally, we assess whether a model estimated using only purchase-stage data can recover the true distribution of consumer types. Although the estimator exhibits only minor bias when the full sample is used, the estimated distribution becomes severely distorted when relying solely on the second half of the sample. This finding highlights the potential bias of using purchase data alone for estimation, particularly given the absence of purchase history prior to the start of our sample period.

C.2 Demand-side: Estimation details

This subsection outlines the construction of the sub-sample used for structural estimation, presents the individual likelihood function based on observed purchases and usage behavior, and describes the selection of R through model fit comparisons.

Structural sample construction. We retain a final sample of $34,548$ active users who exhibited both positive spending and usage throughout the sample period and remained on Steam through at least November 2017. This user sample is used to construct the summary statistics of consumer behavior in Section 3. Summary statistics on seller pricing patterns, also reported in Section 3, are based on the full sample of games.

Due to computational constraints, we draw a sub-sample of users and video games for structural estimation. We focus on capturing competition among games that were popular either for purchases or for usage during the sample period. Thus, we select games that meet one of the following criteria: (1) they are in the top 70% of expenditure share and have more than 1% of users buying among the 34,548 active users, or (2)

Table C.1: Parameter Estimates from Full Sample and Later Half, with and without Usage

	True	Full Sample		Later Half		Full Sample (No usage)		Later Half (No usage)	
Parameter	Value	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.
Usage-game1	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game2	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game3	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game4	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game5	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game6	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game7	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game8	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game9	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
Usage-game10	1.00	1.00	(0.00)	1.00	(0.00)	–	–	–	–
γ	0.60	0.60	(0.00)	0.60	(0.00)	–	–	–	–
σ	1.00	1.0	(0.00)	1.01	(0.01)	–	–	–	–
Purchase-game1	-0.50	-0.47	(0.02)	-0.44	(0.04)	1.64	(0.01)	0.64	(0.05)
Purchase-game2	-0.50	-0.47	(0.02)	-0.43	(0.03)	1.38	(0.01)	0.77	(0.04)
Purchase-game3	-1.00	-0.97	(0.02)	-0.91	(0.03)	0.60	(0.01)	0.56	(0.04)
Purchase-game4	-1.00	-0.99	(0.02)	-0.94	(0.04)	0.70	(0.01)	0.52	(0.05)
Purchase-game5	-1.50	-1.47	(0.02)	-1.42	(0.03)	0.04	(0.01)	0.12	(0.04)
Purchase-game6	-1.50	-1.48	(0.02)	-1.43	(0.04)	0.07	(0.01)	0.18	(0.04)
Purchase-game7	-2.00	-1.98	(0.02)	-1.91	(0.04)	-0.45	(0.02)	-0.26	(0.05)
Purchase-game8	-2.00	-1.98	(0.02)	-1.92	(0.04)	-0.49	(0.02)	-0.37	(0.05)
Purchase-game9	-2.50	-2.49	(0.02)	-2.44	(0.04)	-0.96	(0.01)	-0.89	(0.06)
Purchase-game10	-2.50	-2.51	(0.02)	-2.49	(0.04)	-1.03	(0.02)	-0.96	(0.05)
α	2.00	2.01	(0.01)	2.04	(0.02)	1.68	(0.00)	1.66	(0.02)
Hetero-usage$_{\text{diff}}$	-1.00	-1.00	(0.00)	-1.00	(0.00)	–	–	–	–
Hetero-purchase$_{\text{diff}}$	-1.00	-0.99	(0.04)	-1.02	(0.09)	1.71	(0.04)	1.04	(0.09)
Hetero-alpha$_{\text{diff}}$	2.00	2.01	(0.02)	2.01	(0.04)	2.28	(0.02)	1.56	(0.04)
Fraction of type 1	0.30	0.30	(0.00)	0.30	(0.00)	0.30	(0.00)	0.07	(0.00)

Note: We simulate consumer purchase and usage behavior for 6000 consumers and 240 days, approximating our structural sample. We implement estimation for $MC = 30$ and the number of games equal to 10 to keep the simulation tractable. The type 1 consumers are the heavy type, who take 30% of the market.

they are in the top 70% of timeshare and have more than 1% of players among the active users. After this step, we have a set of 174 popular games. We find 31,571 (91%) of the active users have both purchased and played at least one of the popular games.

Next, we randomly sample 6,314 (20%) of the 31,571 users for structural estimation. We further exclude 3 games that have fewer than 10 purchases or usage sessions among the sampled users. As a result, we include 171 popular games as inside goods (with features and prices tracked), and group all the other games into a fringe bin. On average, the inside games take up 68% of the expenditure and 75% of the playing of a user.

Model comparison and selection. We evaluate model fit across specifications with different values of R using AIC, BIC, and quantiles of individual likelihoods. We find that increasing the number of types from 7 to 8 yields only modest improvements, suggesting limited additional value from assuming more types in the market.

Table C.2: Model Fit Comparison

	AIC	BIC	LL_i (mean)	LL_i (25 pct)	LL_i (median)	LL_i (75 pct)
Model 1	11,711,780	11,716,510	−942	−1,213	−844	−561
Model 2	11,254,573	11,259,736	−917	−1,197	−827	−540
Model 3	11,067,342	11,072,937	−901	−1,176	−823	−533
Model 4	10,970,204	10,976,232	−893	−1,161	−816	−533
Model 5	10,921,130	10,927,590	−889	−1,156	−811	−533
Model 6	10,874,764	10,881,657	−885	−1,153	−813	−524
Model 7	10,842,651	10,849,976	−882	−1,152	−812	−522
Model 8	10,810,874	10,818,631	−879	−1,150	−811	−522

C.3 Supply-side: Solving for the equilibrium prices.

As mentioned in Section 4.2, for each information design, we solve the static pricing equilibrium for each sampling week separately and then aggregate across weeks. This Section describes how we solve for the equilibrium prices.

In week t_w, a consumer's choice depends on her library status L_{i,t_w}. Because our platform information design does not permit targeting within the provided segment, solving for the optimal targeted prices requires an integration over consumers' libraries. We draw consumer library status from the empirical distribution when calculating $\mathbb{E}[s_{jr,t_w}] = \mathbb{E}_i[s_{ji,t_w}(L_{i,t_w})|\boldsymbol{\Theta_r}]$. Specifically, for the counterfactual targeting scenario with information sets $\boldsymbol{\mathcal{I}}$, we compute the equilibrium prices $\boldsymbol{P(\mathcal{I})}$ following three steps.

1. We calculate $v_{i,t_w}(\boldsymbol{\Theta_r}) = \mathbb{E}\Big[V^{usage}_{it_w}(L_{i,t_w-1} \cup \{j\})\Big|\boldsymbol{\Theta_r}\Big]$ for $r \in \{1, ...R\}$,$i \in \{1, ...I\}$.
2. For each type, we have $\mathbb{E}[s_{jr,t_w}] = \mathbb{E}_i\Big[s_{ji,t_w}(v_{i,t_w}(\boldsymbol{\Theta_r}))\Big]$.
3. We substitute s_{jr} in (15) with $\mathbb{E}[s_{jr,t_w}]$ to get the optimal prices.

We implement the procedure for a subset of weeks, specifically $t_w \in \{1, 9, 17, 25, 33, 41\}$, and aggregate profits across these periods. To keep the computation tractable, we use one-tenth of the consumer sample across all information designs. We confirm that this reduced sample yields the same set of equilibrium prices as the full sample, while substantially reducing computational time.

Generalized equilibrium that accommodates asymmetric information sets across sellers. We extend the Bertrand-Nash equilibrium framework in Section 4 to allow for asymmetric information sets across sellers. Let CR^f denote the information set available to seller f, and define $\mathcal{I} = \{CR^1, \ldots, CR^F\}$ as the collection of all sellers' information sets. Given $\mathcal{I}$, each seller $f \in \{1, \ldots, F\}$ solves the profit-maximization problem in equation (19), choosing prices to maximize the expected profits from its product portfolio, J_f.

$$\max_{\{\boldsymbol{P}_j(\mathcal{I})\}_{j \in J_f}} \sum_{j=1}^{J} w_{j,f} \cdot \sum_{cr \in CR^f} p_{j,cr} \cdot \overbrace{\Big(\tilde{\psi}_{cr} \cdot \sum_{r=1}^{R} \tilde{\psi}_{r|cr} \cdot s_{j,r}\big(p_{j,cr}, \boldsymbol{P_{-j}(\mathcal{I})}\big) \Big)}^{\text{expected purchases in coarsened segment } cr}, \tag{19}$$

In the above, $w_{j,f}$,$\tilde{\psi}_{cr}$,$s_{j,r}$ and $s_{j,r}$ are calculated using the same specification as in Equation (14). Asymmetric information sets across sellers enters through the specification of $\mathcal{I}$. Implicitly, this generalized equilibrium concept assumes that all sellers are aware of the information sets held by their competitors.

D Additional results

D.1 Demand-side: Estimates and Interpretations

We present the demand estimates in the following order: homogeneous parameters for the usage and purchase stages, heterogeneous parameters for both stages, and game fixed effects. The left panel of Table D.1 shows that consumers exhibit significant satiation behavior and face lower time costs on weekends, during summer, and over holidays—— consistent with increased gameplay during these periods. In the purchase stage, as shown in the right panel of Table D.1, we find statistically significant positive coefficients for both pre-sale and sale event periods, suggesting these effects are not driven by inventory stockpiling behavior. Additionally, we find that the connection between the two stages, σ, is statistically significant, confirming that consumers take expected future usage into account when making purchase decisions.

Table D.2 presents the game fixed effects with descending order of usage popularity, measured by the observed total number of usage sessions. Two observations stand out. First, games that are played more frequently are estimated to have higher usage-stage fixed effects, suggesting that our usage model does capture revealed consumer preferences. Second, games with higher usage-stage fixed effects are not necessarily estimated to have high purchase-stage fixed effects. The discrepancy in these two-stage fixed effects reflect

Table D.1: Homogeneous parameters

parameter	estimate	s.e.
γ	3.91	0.004
c_{wkd}	-0.03	0.001
c_{smr}	-0.01	0.001
c_{hld}	-0.02	0.002

parameter	estimate	s.e.
ξ_{pre_event}	0.16	0.02
ξ_{sales_event}	0.56	0.01
σ	0.11	0.001

Note: The left panel displays estimates for the usage-stage and the right panel displays those for the purchase stage.

consumers' different considerations when making purchase and usage decisions, underscoring the importance of including two sets of fixed effects in our model.

Table D.2: Game fixed effects

Game	**Usage**	**s.e.**	**Purchase**	**s.e.**
Football Manager 2018	1.00	0.01	-5.51	0.21
Assassin's Creed Origins	0.98	0.01	-3.46	0.20
Middle-earth: Shadow of War	0.94	0.005	-3.03	0.15
Call of Duty: WWII - Multiplayer	0.93	0.01	-5.33	0.13
PLAYERUNKNOWN'S BATTLEGROUNDS	0.90	0.002	-3.94	0.11
Total War: WARHAMMER II	0.90	0.004	-5.50	0.15
Divinity: Original Sin 2	0.89	0.004	-2.62	0.17
Wolfenstein II: The New Colossus	0.88	0.01	-6.65	0.15
fringe	0.87	0.002	-2.25	0.01
Call of Duty: WWII	0.80	0.01	-4.63	0.14
TEKKEN 7	0.76	0.01	-5.84	0.26
Football Manager 2017	0.76	0.004	-6.91	0.18
Dead Cells	0.73	0.01	-6.17	0.12
Cuphead	0.72	0.01	-4.29	0.15
Tom Clancy's Ghost Recon Wildlands	0.72	0.004	-5.02	0.17
Hearts of Iron IV	0.72	0.004	-7.06	0.15
For Honor	0.71	0.004	-5.38	0.14
Counter-Strike: Global Offensive	0.69	0.002	-5.77	0.08
RimWorld	0.69	0.004	-5.83	0.16
Tom Clancy's Rainbow Six Siege	0.68	0.003	-6.18	0.09
Friday the 13th: The Game	0.68	0.01	-6.29	0.16
Oxygen Not Included	0.66	0.01	-6.09	0.14
NieR:Automata	0.66	0.01	-4.77	0.26
Stellaris	0.65	0.004	-5.81	0.13
Europa Universalis IV	0.65	0.004	-7.19	0.16
Grand Theft Auto V	0.64	0.003	-3.26	0.11
Sniper Elite 4	0.64	0.01	-6.25	0.16
Black Desert Online	0.64	0.01	-6.84	0.10
Total War: WARHAMMER	0.63	0.003	-6.28	0.12
Z1 Battle Royale	0.63	0.003	-6.50	0.10
Rocket League	0.63	0.002	-6.77	0.10
Hollow Knight	0.62	0.01	-6.43	0.12
Prey	0.61	0.01	-6.40	0.18
DARK SOULS III	0.61	0.004	-4.63	0.15
Battlerite	0.60	0.004	-5.81	0.08
theHunter: Call of the Wild	0.60	0.01	-8.37	0.16
The Binding of Isaac: Rebirth	0.58	0.004	-7.84	0.16
Stick Fight: The Game	0.58	0.01	-6.43	0.12
Rising Storm 2: Vietnam	0.58	0.01	-7.03	0.12
Warhammer 40,000: Dawn of War III	0.58	0.01	-6.74	0.15
Dota 2	0.57	0.003	-7.69	0.12
Factorio	0.56	0.01	-7.44	0.15
Conan Exiles	0.56	0.01	-6.74	0.15
Dead by Daylight	0.56	0.004	-7.16	0.13
XCOM 2	0.56	0.004	-6.48	0.16
The Witcher 3: Wild Hunt	0.55	0.004	-3.53	0.11
Arma 3	0.54	0.003	-6.14	0.12
Fallout 4	0.53	0.004	-5.21	0.13

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Table D.2 – continued from previous page

Game	Usage	s.e.	Purchase	s.e.
Grim Dawn	0.53	0.01	-7.18	0.13
The Elder Scrolls Online	0.52	0.005	-5.62	0.11
Mount and Blade: Warband	0.52	0.005	-7.01	0.15
Bayonetta	0.52	0.02	-6.97	0.18
Blackwake	0.51	0.01	-8.27	0.15
Sid Meier's Civilization VI	0.51	0.004	-5.91	0.18
ARK: Survival Evolved	0.50	0.004	-5.39	0.12
Stardew Valley	0.50	0.005	-6.30	0.11
Darkest Dungeon	0.50	0.01	-5.95	0.12
Euro Truck Simulator 2	0.50	0.005	-6.57	0.12
7 Days to Die	0.49	0.01	-6.99	0.14
Shadow Tactics: Blades of the Shogun	0.49	0.02	-6.17	0.16
Warframe	0.46	0.004	-6.22	0.08
Northgard	0.46	0.01	-7.50	0.13
Rust	0.45	0.004	-6.67	0.12
Human: Fall Flat	0.44	0.02	-7.49	0.15
Killing Floor 2	0.43	0.01	-6.50	0.14
Paladins	0.42	0.01	-6.81	0.10
Kingdoms and Castles	0.41	0.02	-7.08	0.13
Path of Exile	0.41	0.004	-6.78	0.08
The Elder Scrolls V: Skyrim Special Edition	0.41	0.005	-5.75	0.14
Dishonored 2	0.41	0.01	-6.27	0.17
Terraria	0.41	0.004	-6.89	0.12
DARK SOULS II: Scholar of the First Sin	0.40	0.01	-5.80	0.14
Cities: Skylines	0.40	0.01	-6.55	0.12
Dying Light	0.39	0.01	-6.77	0.14
The Elder Scrolls V: Skyrim	0.39	0.004	-6.44	0.15
Mortal Kombat X	0.39	0.01	-7.60	0.14
Garry's Mod	0.38	0.004	-6.70	0.11
Golf With Your Friends	0.38	0.01	-7.67	0.12
Middle-earth: Shadow of Mordor	0.37	0.01	-5.43	0.08
Sid Meier's Civilization V	0.37	0.004	-6.76	0.13
Subnautica	0.36	0.01	-6.46	0.14
The Forest	0.35	0.01	-6.70	0.13
Just Cause 3	0.34	0.01	-5.15	0.13
Overcooked	0.34	0.02	-7.99	0.19
Planet Coaster	0.34	0.01	-4.88	0.20
Sleeping Dogs: Definitive Edition	0.33	0.01	-6.97	0.14
DOOM	0.33	0.01	-5.74	0.12
Rise of the Tomb Raider	0.33	0.01	-4.84	0.15
Borderlands 2	0.33	0.01	-6.52	0.14
Starbound	0.32	0.01	-7.21	0.14
Team Fortress 2	0.31	0.005	-7.69	0.14
PAYDAY 2	0.31	0.004	-4.87	0.07
Slime Rancher	0.30	0.01	-7.29	0.15
Wolfenstein: The New Order	0.30	0.01	-7.54	0.12
Warhammer: End Times - Vermintide	0.30	0.01	-7.29	0.12
SMITE	0.29	0.01	-7.81	0.12
The Long Dark	0.29	0.01	-7.03	0.19
Deus Ex: Mankind Divided	0.29	0.01	-7.53	0.17
Offworld Trading Company	0.28	0.02	-7.71	0.17
DiRT Rally	0.27	0.01	-7.17	0.17
Mad Max	0.27	0.01	-7.86	0.17
Tabletop Simulator	0.26	0.01	-6.98	0.14
War Thunder	0.26	0.01	-7.33	0.10
Batman: Arkham Knight	0.25	0.01	-7.11	0.16
Insurgency	0.24	0.01	-7.14	0.09
Don't Starve Together	0.22	0.01	-6.43	0.11
Warhammer 40,000: Eternal Crusade	0.22	0.02	-7.85	0.12
Age of Empires II (2013)	0.22	0.01	-7.40	0.15
Verdun	0.22	0.01	-8.55	0.19
SUPERHOT	0.20	0.02	-6.97	0.14
Move or Die	0.20	0.02	-7.86	0.18
Ultimate Chicken Horse	0.20	0.02	-7.94	0.20
Ori and the Blind Forest: Definitive Edition	0.20	0.02	-6.20	0.14
Borderlands: The Pre-Sequel	0.20	0.01	-6.11	0.19
Poly Bridge	0.20	0.02	-8.41	0.21
Hyper Light Drifter	0.19	0.02	-7.43	0.16
SimplePlanes	0.18	0.03	-9.13	0.24
Kingdom: New Lands	0.17	0.02	-8.27	0.21
INSIDE	0.15	0.02	-7.25	0.15
The Witness	0.15	0.02	-5.68	0.15
Plague Inc: Evolved	0.15	0.02	-8.35	0.15

Continued on next page

Table D.2 – continued from previous page

Game	Usage	s.e.	Purchase	s.e.
Sid Meier's Civilization III: Complete	0.15	0.01	-6.64	0.09
Titan Quest Anniversary Edition	0.14	0.01	-7.66	0.14
Dungeons 2	0.14	0.02	-5.60	0.11
Left 4 Dead 2	0.14	0.01	-6.25	0.10
Hand of Fate	0.12	0.02	-6.93	0.14
This War of Mine	0.11	0.01	-7.22	0.12
Unturned	0.10	0.01	-7.06	0.10
Sniper Elite 3	0.08	0.02	-7.79	0.17
Rising Storm/Red Orchestra 2 Multiplayer	0.08	0.01	-6.81	0.10
South Park: The Stick of Truth	0.08	0.02	-6.05	0.17
Saints Row IV	0.06	0.01	-6.64	0.13
Metro 2033 Redux	0.06	0.01	-7.68	0.13
Lords Of The Fallen	0.06	0.03	-7.45	0.16
SpeedRunners	0.06	0.02	-8.76	0.22
Borderlands GOTY	0.06	0.01	-7.22	0.15
Beholder	0.06	0.03	-7.81	0.15
Alien: Isolation	0.03	0.02	-7.77	0.17
Undertale	-0.002	0.02	-7.54	0.16
Outlast	-0.01	0.02	-5.41	0.08
Layers of Fear	-0.01	0.04	-7.78	0.19
Portal 2	-0.01	0.01	-5.60	0.10
Metro: Last Light Redux	-0.05	0.02	-7.89	0.14
Ashes of the Singularity: Escalation	-0.06	0.06	-6.04	0.17
Killing Floor	-0.06	0.01	-5.76	0.11
Mirage: Arcane Warfare	-0.07	0.04	-5.54	0.09
Chivalry: Medieval Warfare	-0.07	0.01	-5.74	0.10
Shadow Warrior	-0.08	0.02	-5.62	0.09
Galactic Civilizations III	-0.09	0.04	-7.69	0.18
ABZ	-0.09	0.04	-7.04	0.17
RIVE	-0.10	0.06	-7.81	0.18
Fallout	-0.12	0.03	-5.89	0.07
The Witcher 2: Assassins of Kings Enhanced Edition	-0.13	0.02	-6.14	0.11
Tomb Raider	-0.14	0.02	-7.05	0.14
Killer is Dead	-0.16	0.05	-6.15	0.09
The Walking Dead	-0.19	0.03	-6.48	0.15
BioShock Infinite	-0.21	0.02	-6.72	0.14
Jotun: Valhalla Edition	-0.22	0.03	-5.24	0.08
Outland	-0.22	0.03	-6.78	0.10
Brtal Legend	-0.24	0.04	-6.27	0.10
Left 4 Dead	-0.27	0.02	-8.07	0.15
Sanctum 2	-0.28	0.03	-6.25	0.09
Starpoint Gemini 2	-0.31	0.04	-5.52	0.12
BioShock Remastered	-0.34	0.02	-7.85	0.19
Serious Sam Fusion 2017 (beta)	-0.34	0.04	-5.31	0.07
The Bureau: XCOM Declassified	-0.35	0.05	-5.85	0.08
Guns of Icarus Online	-0.37	0.03	-7.22	0.10
Portal	-0.51	0.02	-7.35	0.14
Saints Row 2	-0.53	0.04	-6.25	0.09
Psychonauts	-0.58	0.06	-7.53	0.16
Monaco	-0.59	0.06	-9.26	0.25
Alan Wake's American Nightmare	-1.02	0.09	-7.10	0.13

Moreover, Tables D.3 and D.4 reveal notable discrepancies across consumer types, particularly in overall engagement. For example, Type 7 consumers generally enjoy playing games but are less likely to make purchases and are more price sensitive compared to other types. We also observe meaningful variation in preferences for specific game features.

Interpreting $\hat{\delta}_j$, $\hat{\boldsymbol{\lambda}}$, and $\hat{\boldsymbol{\alpha}}_{\boldsymbol{r}}$. Although we do not fully disentangle the underlying channels that are discussed in Section 4.2, we explore whether consumer learning is reflected in $\hat{\delta}_j$, $\hat{\boldsymbol{\lambda}}_{\boldsymbol{r}}$, and $\hat{\boldsymbol{\alpha}}_{\boldsymbol{r}}$. Figure D.1 plots each game's purchase-stage fixed effect against its usage-stage fixed effect. If post-purchase learning drove the purchase-stage effect, we would expect games with low usage (and thus more learning potential)

Table D.3: Usage-stage heterogeneity ($\boldsymbol{\lambda}$)

Parameter	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8
fraction	14%	11%	12%	8%	22%	6%	8%	19%
any	0	-0.6	0.3	0.1	-0.4	-0.5	0.1	-0.6
	-	(0.008)	(0.006)	(0.005)	(0.007)	(0.008)	(0.006)	(0.009)
multiplayer	0	-0.07	-0.07	0.07	-0.07	0.14	0.10	0.11
	-	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)	(0.006)
military	0	0.08	-0.09	-0.05	-0.05	0.08	-0.03	-0.06
	-	(0.004)	(0.004)	(0.003)	(0.004)	(0.003)	(0.003)	(0.006)
survival	0	0.22	-0.14	0.06	-0.01	-0.09	0.03	-0.03
	-	(0.004)	(0.002)	(0.002)	(0.004)	(0.003)	(0.002)	(0.004)
team	0	0.24	0.05	0.11	-0.12	0.20	-0.15	0.07
	-	(0.004)	(0.004)	(0.003)	(0.005)	(0.004)	(0.003)	(0.006)
fantasy	0	0.33	-0.17	0.13	-0.05	0.24	-0.06	-0.23
	-	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.003)	(0.004)
platformer	0	0.01	-0.06	-0.15	0.04	0.10	-0.12	0.21
	-	(0.004)	(0.002)	(0.002)	(0.004)	(0.003)	(0.002)	(0.004)
dark	0	-0.038	-0.049	-0.037	-0.070	0.004	0.087	-0.209
	-	(0.004)	(0.004)	(0.003)	(0.005)	(0.004)	(0.003)	(0.006)
adventure	0	-0.08	-0.09	-0.05	-0.01	0.32	-0.12	-0.19
	-	(0.006)	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)	(0.008)
anime	0	-0.10	-0.11	-0.31	0.04	0.06	0.05	-0.33
	-	(0.006)	(0.006)	(0.007)	(0.007)	(0.005)	(0.005)	(0.012)
fringe	0	-0.01	-0.53	-0.18	-0.11	0.36	-0.34	-0.57
	-	(0.008)	(0.006)	(0.005)	(0.007)	(0.007)	(0.005)	(0.010)

to exhibit higher purchase-stage fixed effects, reflecting the overestimation of game quality at the point of purchase. Instead, we observe only a weak overall correlation and a slight positive correlation among the most frequently played games. This pattern is more consistent with Steam's recommendation algorithm, which amplifies popular games in consideration.

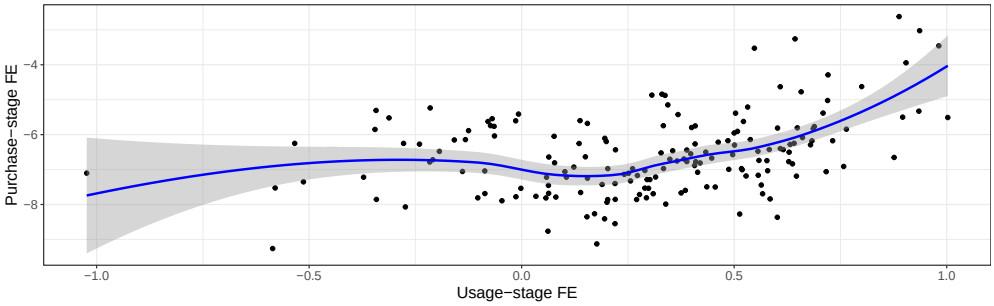


Figure D.1: Correlation between two-stage FEs: η_j and δ_j

Additionally, across consumer types, we find that stronger feature preferences correlate weakly with more negative price sensitivities. This pattern again aligns with algorithmic recommendations, which amplifies games matching individual tastes in the purchase-stage consideration, rather than with learning behavior about potential match value with the features. Taken together, the pattern of estimates $\hat{\delta}_j$, $\hat{\boldsymbol{\lambda}}_{\boldsymbol{r}}$, and $\hat{\boldsymbol{\alpha}}_{\boldsymbol{r}}$ is closely aligned with the recommendation algorithm instead of consumer learning.

Table D.4: Purchase-stage heterogeneity

(a) Intercepts for meta-tags ($\boldsymbol{\beta}$)

Parameter	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8
any	0	-1.4	-0.8	-0.8	-0.6	-0.5	-0.8	-2.0
	-	(0.2)	(0.1)	(0.2)	(0.1)	(0.2)	(0.2)	(0.2)
multiplayer	0	0.40	0.17	0.39	-0.09	-0.11	0.40	0.45
	-	(0.11)	(0.09)	(0.12)	(0.08)	(0.11)	(0.11)	(0.10)
military	0	-0.19	0.13	0.05	-0.05	0.10	0.13	0.19
	-	(0.11)	(0.09)	(0.11)	(0.08)	(0.11)	(0.10)	(0.10)
survival	0	0.2	0.5	0.3	0.2	0.1	0.4	0.5
	-	(0.10)	(0.08)	(0.10)	(0.07)	(0.11)	(0.09)	(0.09)
team	0	0.32	0.35	0.30	0.05	0.09	0.29	0.25
	-	(0.12)	(0.09)	(0.12)	(0.08)	(0.12)	(0.10)	(0.10)
fantasy	0	-0.23	-0.31	-0.39	-0.08	-0.05	-0.19	-0.21
	-	(0.10)	(0.08)	(0.10)	(0.07)	(0.11)	(0.10)	(0.09)
platformer	0	0.25	0.16	0.33	0.14	0.15	0.03	0.23
	-	(0.10)	(0.08)	(0.10)	(0.07)	(0.11)	(0.09)	(0.09)
dark	0	-0.08	-0.06	-0.05	-0.15	-0.11	-0.08	-0.26
	-	(0.10)	(0.08)	(0.10)	(0.07)	(0.11)	(0.10)	(0.09)
adventure	0	0.04	0.02	-0.08	-0.01	-0.06	0.19	0.01
	-	(0.12)	(0.10)	(0.12)	(0.08)	(0.11)	(0.11)	(0.11)
anime	0	0.07	0.27	0.20	0.19	0.02	0.29	0.50
	-	(0.2)	(0.1)	(0.2)	(0.1)	(0.2)	(0.1)	(0.1)
fringe	0	0.11	0.16	0.23	-0.10	-0.05	0.24	0.39
	-	(0.2)	(0.1)	(0.2)	(0.1)	(0.2)	(0.2)	(0.2)

(b) Price sensitivities ($\boldsymbol{\alpha}$)

Parameter	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8
any	-0.070	-0.073	-0.080	-0.088	-0.074	-0.069	-0.094	-0.047
	(0.0050)	(0.0107)	(0.0089)	(0.0112)	(0.0079)	(0.0104)	(0.0102)	(0.0096)
multiplayer	0.0207	0.0012	0.0220	0.0122	0.0187	0.0271	0.0170	0.0031
	(0.0026)	(0.0052)	(0.0047)	(0.0057)	(0.0040)	(0.0052)	(0.0054)	(0.0050)
military	0.034	0.046	0.027	0.033	0.034	0.021	0.023	0.026
	(0.0027)	(0.0059)	(0.0048)	(0.0058)	(0.0042)	(0.0054)	(0.0053)	(0.0053)
survival	0.00171	0.00075	-0.00857	-0.00576	-0.00307	0.00192	-0.00268	-0.01178
	(0.0025)	(0.0052)	(0.0044)	(0.0053)	(0.0039)	(0.0051)	(0.0050)	(0.0047)
team	-0.00352	-0.00600	-0.00579	-0.00536	-0.00034	0.00059	0.00019	-0.00342
	(0.0027)	(0.0055)	(0.0048)	(0.0059)	(0.0043)	(0.0056)	(0.0054)	(0.0050)
fantasy	-0.0072	-0.0128	-0.0068	-0.0039	-0.0088	-0.0144	-0.0039	-0.0270
	(0.0027)	(0.0054)	(0.0046)	(0.0057)	(0.0041)	(0.0054)	(0.0053)	(0.0049)
platformer	-0.022	-0.032	-0.024	-0.027	-0.027	-0.031	-0.012	-0.036
	(0.0024)	(0.0049)	(0.0042)	(0.0050)	(0.0037)	(0.0048)	(0.0046)	(0.0045)
dark	0.0089	0.0163	0.0193	0.0205	0.0152	0.0163	0.0193	0.0279
	(0.0026)	(0.0054)	(0.0046)	(0.0056)	(0.0041)	(0.0054)	(0.0053)	(0.0049)
adventure	0.01017	-0.00069	0.00132	-0.00290	0.00579	0.00387	-0.01161	-0.00628
	(0.0027)	(0.0059)	(0.0049)	(0.0065)	(0.0041)	(0.0052)	(0.0062)	(0.0054)
anime	-0.0018	-0.0124	-0.0178	-0.0307	-0.0073	-0.0039	-0.0210	-0.0426
	(0.0049)	(0.0085)	(0.0078)	(0.0106)	(0.0066)	(0.0085)	(0.0091)	(0.0088)

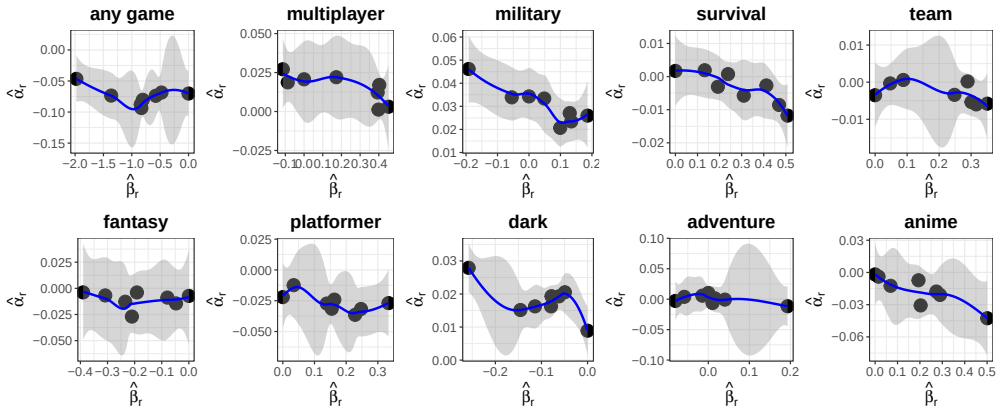


Figure D.2: Correlation between purchase tastes and sensitivity: β_r and α_r

Implied price elasticities and substitution: additional details. Figure 5.4 reveals substantial heterogeneity in consumers' own-price elasticities across games. One concern is that using the raw elasticities as a similarity metric may overweight discrepancies in games with larger dispersion in elasticities, thereby impacting the effectiveness of the resulting segmentation in increasing profits. Therefore, Figure D.3 visualizes a rescaled elasticity pattern across consumers toward games. For each game j, we compute $\tilde{\eta}jj^r = -\eta jj^r / \max_r |\eta^r_{jj}|$ so that the most price-elastic type for each game has a relative elasticity of 1 and all others lie between 0 and 1. This normalization adjusts for game-specific differences in price sensitivity and reveals how the ranking of "strong" versus "weak" markets varies by game title. In Section 6, we use the rescaled elasticity measure as an alternative metric to capture demand similarity across consumer types.

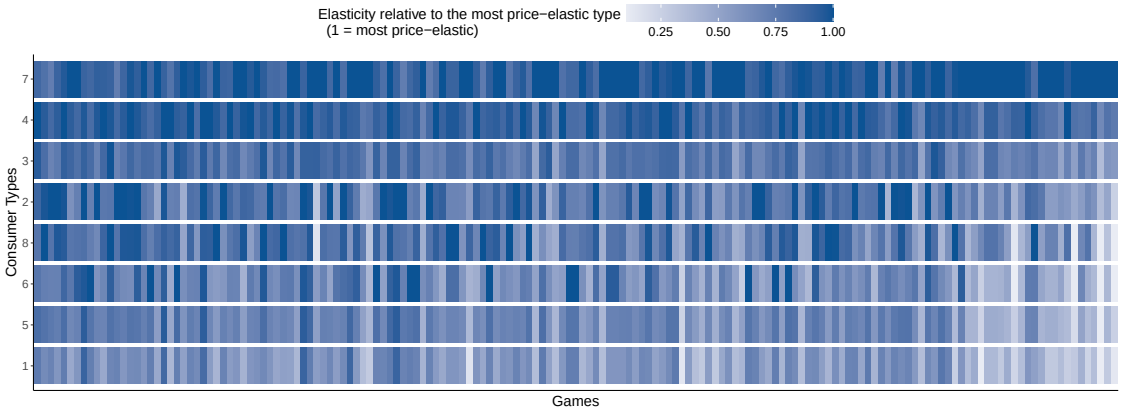


Figure D.3: Rescaled own elasticities across games and consumer types.

Note: We evaluate all price elasticities at the observed prices. For visualization, we re-scale game-type price elasticities by computing $\tilde{\eta}^r_{jj} = \eta^r_{jj} / \max_r |\tilde{\eta}^r_{jj}|$. The darker a tile is, the more price-sensitive the type is towards the game. We sort the types by their mean elasticity (across games) and sort the games by their market-level elasticities.

In the meantime, Table D.5 highlights heterogeneity in substitution patterns across consumer types. For each type, we report the average diversion ratio from a focal game to all other inside goods. The results show

meaningful variation in substitution behavior, most strikingly among Type 1 consumers, the most engaged type, who shift a large share of their purchases to alternative games when a focal title is removed.

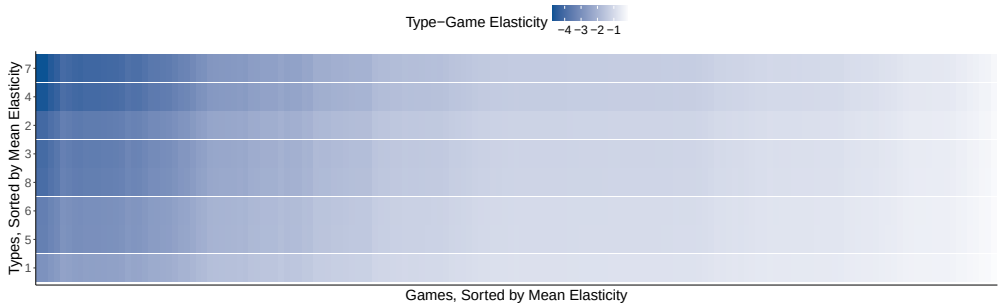
Table D.5: Diversion ratios to inside goods: across consumer types

	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8
Fraction	14%	11%	12%	8%	22%	6%	8%	19%
Diversion Ratio	0.28	0.09	0.16	0.16	0.16	0.19	0.17	0.07

D.2 Supply-side: Generalization

Alternative consumer heterogeneity patterns. Figure D.4 illustrates the two distinct heterogeneity patterns uncovered in the decomposition exercise (Section 6.3), while Table D.6 compares the coarse segmentation formed using optimal information design and the demand-based heuristics under the two counterfactual demand scenarios.

(a) Vertical heterogeneity: elasticity toward games



(b) Horizontal heterogeneity: elasticity toward games

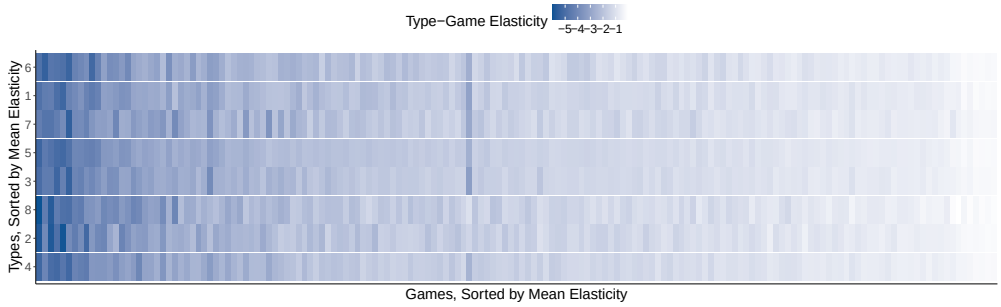


Figure D.4: Alternative demand: vertical versus horizontal heterogeneity

Evidently, even under one-dimensional heterogeneity, profit gaps persist across varying levels of granularity. Specifically, when consumer differences are purely vertical (i.e., engagement and price sensitivity)—that is, a scenario where the sellers are aligned in targeting "strong" and "weak" markets, the elasticity-based heuristics fail to approximate the optimal design. Similar to Section 6.2, clustering based on vertical similarity alone isolates the most distinct type, whereas the profit-maximizing design instead pools that type with others. Moreover, comparison between Table 6.1 and Table D.6 suggests that the optimal "Coarse 2"

Table D.6: Heterogeneity decomposition: Vertical versus Horizontal

(a) Vertical (full-info profit gain: 1.73%)

Coarse-n	Heuristic	Profit gain	Optimal	Profit gain
$n = 2$	{1,2,3,4,5,6,8},{7}	0.36%	{1,5,6},{2,3,4,7,8}	1.23%
$n = 7$	{1,5},{2},{3},{4},{6},{7},{8}	1.40%	{2,8},{1},{3},{4},{5},{6},{7}	1.73%

(b) Horizontal (full-info profit gain: 0.96%)

Coarse-n	Heuristic	Profit gain	Optimal	Profit gain
$n = 2$	{1,2,3,4,5,6,7},{8}	0.30%	{2,4,7},{1,3,5,6,8}	0.43%
$n = 7$	{1,7},{2},{3},{4},{5},{6},{8}	0.88%	{5,6},{1},{2},{3},{4},{7},{8}	0.94%

design under both dimensions of heterogeneity (6.2) reflects pooling in the vertical distinction, while the finer optimal "Coarse 7" reflects pooling in the horizontal distinction.

Other objectives: consumer welfare by type. Table D.7 reports changes in consumer welfare by type. Relative to the uniform-pricing benchmark, total welfare gains arise primarily from lower prices for the more price-elastic types. Moreover, the optimal "Coarse 2" segmentation delivers larger welfare improvements than the heuristic "Coarse 2" by implementing price cuts that benefit a greater share of consumers, while imposing only modest price increases on the remainder of the market.

Table D.7: Changes in Price and Consumer Surplus by Type across Information Designs

	Heuristic Coarse-2		Optimal Coarse-2		Full-info	
Type	ΔPrice(%)	ΔCS	ΔPrice(%)	ΔCS	ΔPrice(%)	ΔCS
1	−5.0	−4	5.5	−26	23.9	−47
2	−5.0	−4	−21.8	32	−8.7	31
3	−5.0	−6	−21.8	38	−10.9	27
4	−5.0	−3	−21.8	25	−23.4	42
5	−5.0	−8	5.5	−50	6.2	−18
6	−5.0	−2	5.5	−14	7.9	−3
7	−31.0	42	−21.8	28	−25.6	42
8	−5.0	−8	−21.8	59	−3.1	42